

LIFE SCIENCES PRE-SERVICE TEACHERS' PERCEPTIONS AND PREPAREDNESS FOR INTEGRATION OF AI IN ONE SOUTH AFRICAN UNIVERSITY

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DECLARATION

I hereby declare that this research report is my own unaided work.
It is submitted for the degree of Masters of Education in Science Education at the University of the Witwatersrand. It has not been submitted before for any other degree or examination at any university.

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ABSTRACT

This study explores Life Sciences pre-service teachers' perceptions and preparedness for the integration of Artificial Intelligence (AI) within Initial Teacher Education (ITE). As AI continues to reshape educational practices, there is a growing need to understand how future teachers are being prepared to engage with these technologies in subject-specific contexts. This qualitative exploratory case study, grounded in an interpretivist paradigm, involved 38 pre-service teachers and employed open-ended questionnaires and semi-structured interviews to generate data. The data were analyzed using thematic analysis, informed by the Technology Acceptance Model and the Technological Pedagogical Content Knowledge framework. The findings revealed that participants demonstrated basic and largely self-directed familiarity with AI tools, accompanied by positive perceptions of AI's potential to support the teaching of complex Life Sciences concepts, particularly through visualization and simplification. However, a clear knowledge–practice gap emerged, as participants reported limited pedagogical training and uncertainty regarding effective classroom integration. Participants also expressed ethical concerns and highlighted contextual constraints, including infrastructural limitations, which may influence implementation. The study concludes that while pre-service teachers show strong perceived usefulness of AI, their preparedness for pedagogical integration remains constrained by insufficient formal preparation within teacher education programs. The study recommends the integration of structured, subject-specific AI training within Initial Teacher Education to support meaningful, context-responsive adoption.

Keywords: Artificial Intelligence, Pre-service Teachers, Life Sciences Education, AI Integration, Teaching Preparedness, Digital Pedagogy, South African Classrooms.

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CHAPTER ONE

CONTEXT OF THE STUDY

1.1 Introduction

Artificial Intelligence, often referred to as AI, is broadly defined as the ability of machines or computer systems to simulate human intelligence through learning, reasoning, problem-solving, and decision-making (Luckin et al., 2016), has become an integral part of the fourth industrial revolution (4IR) (Chen et al., 2020). Globally, AI-driven tools such as intelligent tutoring systems, generative platforms, and adaptive learning environments are being integrated into classroom practices, prompting renewed discussions about teacher preparedness and digital competence (Ertmer & Ottenbreit-Leftwich, 2010; Viberg et al., 2023). While much of this discourse focuses on schooling contexts and in-service teachers, comparatively less attention has been paid to how AI is positioned within Initial Teacher Education (ITE) programmes, where future educators are professionally formed (Mnguni, 2023; Lee & Zhai, 2024; Jeong & Kwon, 2026).

ITE institutions serve as critical sites for shaping teacher identity, pedagogical beliefs, and technological orientations (Holmes & Tuomi, 2022). Research consistently demonstrates that teachers' beliefs, perceptions, and confidence significantly influence technology adoption, often more strongly than infrastructure or policy mandates (Ertmer & Ottenbreit-Leftwich, 2010). As such, understanding how pre-service teachers perceive and experience emerging technologies during their training is essential, since these early professional encounters frequently shape later classroom practice (Mnguni, 2024).

Within the South African context, digital transformation efforts are unfolding alongside persistent infrastructural inequalities, uneven resource distribution, and curriculum pressures (Jere et al., 2023; Tarisayi, 2024). Although national discourse increasingly references the Fourth Industrial Revolution and digital innovation, the extent to which AI is meaningfully integrated into teacher preparation programmes remains unclear (Mnguni, 2023). Existing studies tend to focus on digital technology integration broadly, rather than AI specifically, and frequently prioritise in-service teacher contexts over pre-service populations (Luckin et al., 2016). This creates a significant knowledge gap regarding how South African pre-service

teachers currently experience, interpret, and develop preparedness for AI integration within their ITE programmes.

The discipline of Life Sciences presents a particularly compelling context for this investigation. Life Sciences teaching often involves abstract, microscopic, and complex biological processes that are conceptually challenging for learners (Zawacki-Richter et al., 2019). Emerging AI tools offer possibilities for simulation, modelling, and visualization that may support conceptual understanding in such domains (Holmes et al., 2019). However, the pedagogical value of these tools is mediated by teacher knowledge, beliefs, and perceived competence (Mnguni, 2024). Thus, it becomes critical to explore not merely the potential of AI, but how pre-service Life Sciences teachers within ITE programmes currently perceive their preparedness, exposure, and capacity to integrate such technologies meaningfully.

Rather than positioning pre-service teachers solely as future implementers of AI, this study situates them as present participants within structured teacher education programmes, whose experiences, exposure, and perceptions shape their emerging professional identities (Mnguni, 2023). By foregrounding teacher cognition and preparedness within ITE, the study responds to international scholarship emphasizing that successful technological integration depends fundamentally on teacher beliefs, confidence, and contextualized preparation (Luckin et al., 2016; Zawacki-Richter et al., 2019; Holmes et al., 2019). In doing so, it addresses an underexplored area in South African research concerning AI-specific preparedness among pre-service Life Sciences teachers.

1.2 Background

The integration of emerging technologies into education has long been mediated by teacher beliefs, knowledge structures, and contextual preparation (Teo, 2011). Research consistently demonstrates that technological innovation alone does not guarantee pedagogical transformation; rather, teacher cognition, confidence, and perceived preparedness are decisive factors influencing whether and how technologies are meaningfully adopted (Ertmer & Ottenbreit-Leftwich, 2010; Mnguni, 2023; Mnguni, 2024). Consequently, Initial Teacher Education (ITE) institutions occupy a critical position in shaping how future educators conceptualize, evaluate, and integrate digital tools into their disciplinary practices.

Internationally, ITE programmes are increasingly expected to prepare pre-service teachers not only with subject knowledge and pedagogical competence, but also with technological fluency aligned to evolving digital ecologies (Chai et al., 2013). However, scholarship suggests that technology preparation within teacher education often remains uneven, fragmented, or insufficiently embedded within disciplinary contexts (Adelana et al., 2024; Abualrob, 2025; Zorlu, 2025). Where emerging technologies such as AI are concerned, the literature indicates that teacher education institutions frequently lag behind technological advancement, resulting in uncertainty, limited exposure, and conceptual ambiguity among pre-service teachers (Tondeur et al., 2017).

In the South African context, the integration of digital technologies occurs within a landscape characterized by systemic inequalities, infrastructural disparities, and uneven access to digital resources (Jere et al., 2023; Tarisayi, 2024). While national discourse increasingly emphasizes digital transformation and Fourth Industrial Revolution competencies, there is limited empirical insight into how AI specifically is positioned within South African ITE curricula. Existing studies predominantly focus on general ICT integration or examine in-service teachers' technology adoption, leaving a notable gap regarding AI-specific preparation among pre-service cohorts (Mnguni, 2023; Viberg et al., 2023).

Moreover, teacher education does not merely transmit technical skills; it shapes professional identity and pedagogical orientation (Chai et al., 2013). Pre-service teachers' perceptions of technology, their confidence in using it, and their understanding of its pedagogical alignment are formed within the ITE environment. If AI is not meaningfully introduced, critically examined, or pedagogically contextualized during this formative stage, preparedness for future classroom integration may remain superficial or inconsistent (Howard et al., 2022).

The discipline of Life Sciences further complicates this landscape. Life Sciences teaching involves abstract and microscopic processes that require visualization, modelling, and conceptual scaffolding (Teo, 2011). AI-powered simulations and generative tools hold potential to support such pedagogical needs; however, their effective use depends on teachers' disciplinary knowledge, technological confidence, and contextual awareness (Ertmer & Ottenbreit-Leftwich, 2010). Understanding how pre-service Life Sciences teachers within a

South African ITE programme currently experience, interpret, and assess their preparedness for AI integration is therefore both timely and necessary.

By situating AI preparedness within the context of Initial Teacher Education rather than solely within schooling systems, this study foregrounds the formative environment in which technological beliefs and professional dispositions are shaped. It responds to international and local calls for deeper examination of teacher cognition and preparedness in relation to emerging technologies, particularly within under-researched African contexts (Mnguni, 2024).

1.3 Problem Statement

International scholarship increasingly recognizes that the successful integration of emerging technologies in education is less dependent on the availability of tools and more dependent on teachers' beliefs, perceptions, and knowledge structures (Tomažič et al., 2025; Ishmuradova et al., 2025). Studies conducted in Europe, Asia, and North America suggest that pre-service teachers often demonstrate curiosity toward AI, yet simultaneously report uncertainty regarding pedagogical application, ethical implications, and disciplinary alignment (Ng et al., 2021; Zhai et al., 2021). Despite this growing body of international work, AI-specific research within ITE remains comparatively limited and unevenly distributed across global contexts (Mnguni, 2024).

Within the African context, and particularly in South Africa, empirical research examining AI integration in teacher education is sparse. Existing studies predominantly explore general ICT adoption or focus on in-service teachers' technology practices (Jere et al., 2023). Consequently, little is known about how pre-service teachers, during their formative professional preparation, perceive AI, understand its pedagogical affordances, or evaluate their preparedness to integrate it within subject-specific contexts. This represents a significant research gap, particularly given that beliefs and knowledge developed during ITE often shape future classroom practices (Zhai et al., 2021).

The decision to focus specifically on perceptions and knowledge preparedness is theoretically grounded. Perceptions influence behavioral intention, openness to innovation, and eventual classroom implementation (Tondeur et al., 2017). Teachers who perceive technologies as pedagogically valuable and aligned with disciplinary goals are more likely to adopt them

meaningfully. Conversely, uncertainty, limited exposure, or perceived misalignment can result in resistance or superficial usage (Howard et al., 2022; Viberg et al., 2023). Perceptions therefore function not merely as attitudes, but as cognitive filters through which technological innovations are interpreted.

Similarly, knowledge preparedness extends beyond technical familiarity; it encompasses conceptual understanding, pedagogical integration capacity, and contextual awareness. Without sufficient knowledge preparedness, technological adoption may lack depth, critical engagement, or disciplinary coherence (Mnguni, 2024). Within Life Sciences education, where complex biological systems, abstract processes, and visual representations are central, AI tools may offer substantial pedagogical potential. However, the realization of this potential depends on teachers' subject-specific knowledge structures and their ability to evaluate AI critically within curriculum constraints (Ng et al., 2021).

Accordingly, the purpose of this study is to explore Life Sciences pre-service teachers' perceptions of Artificial Intelligence and to examine their knowledge preparedness for AI integration within a South African Initial Teacher Education programme. By foregrounding teacher cognition within ITE rather than focusing solely on schooling implementation, the study aims to contribute empirical insight into an under-researched area of African educational technology scholarship and to inform curriculum development within teacher preparation programmes.

1.4 Purpose Statement

The purpose of this qualitative exploratory case study is to investigate the perceptions and preparedness of third and fourth-year Life Sciences pre-service teachers within a South African ITE programme regarding the integration of AI as a pedagogical tool. Guided by the Technology Acceptance Model (TAM) and the Technological Pedagogical Content Knowledge (TPACK) framework, the study seeks to examine how these pre-service teachers currently understand, interpret, and experience AI within their teacher preparation programme.

Rather than focusing solely on anticipated future classroom implementation, this study foregrounds the participants' present exposure to AI, their perceived preparedness to integrate AI into subject-specific teaching, and the ways in which their ITE experiences shape their

confidence, beliefs, and conceptual understanding of AI in education. Particular attention is given to how disciplinary knowledge in Life Sciences intersects with emerging technological tools, and how pre-service teachers make sense of this intersection within the constraints and opportunities of their training context.

Ultimately, the study aims to generate empirically grounded insights into how AI preparedness is constructed during initial teacher preparation, thereby contributing to discussions on curriculum design and pedagogical support within ITE programmes in South Africa.

1.5 Significance of the Study

As AI increasingly reshapes educational discourse globally, scholarly attention has shifted toward understanding not only technological affordances, but also the human factors that mediate successful integration. International research consistently demonstrates that teachers' beliefs, perceptions, and knowledge structures function as critical gatekeepers in the adoption of emerging technologies (Ramnarain et al., 2025). While infrastructure and policy frameworks provide enabling conditions, technology integration ultimately depends on how teachers interpret, value, and understand these tools within their pedagogical contexts.

Within the field of science education, AI-driven tools such as intelligent tutoring systems, adaptive simulations, and virtual laboratories are increasingly positioned as transformative innovations (Zhai et al., 2021; Mnguni, 2024). However, much of the existing literature treats science as a unified domain or focuses predominantly on in-service teachers and general ICT usage (Mpungose, 2020; Hlalele, 2021; Mhlanga & Moloi, 2020). This generalization obscures the subject-specific pedagogical demands of Life Sciences, where abstract, systemic, and microscopic processes require specialized representational and instructional strategies (Cochrane & Antonczak, 2021). By focusing specifically on Life Sciences pre-service teachers, this study contributes disciplinary nuance to AI-in-education scholarship and addresses a gap in research on subject-specific technology integration during ITE.

Importantly, AI preparedness in this study is conceptualized not merely as future classroom preparedness, but as a construct shaped within the ITE environment itself. It encompasses pre-service teachers' current exposure to AI tools, their opportunities for pedagogical modelling, their understanding of ethical and contextual implications, and their developing confidence to critically evaluate AI within curriculum constraints. International scholarship suggests that

professional identities, pedagogical beliefs, and technological dispositions formed during teacher preparation often influence later classroom practices (Ertmer & Ottenbreit-Leftwich, 2010; Goksel & Bozkurt, 2019). Investigating preparedness at this formative stage is therefore essential for understanding long-term integration trajectories.

Moreover, global studies indicate that even in contexts with advanced technological infrastructure, teacher uncertainty, limited conceptual understanding, and lack of pedagogical alignment frequently hinder meaningful AI adoption (Viberg et al., 2023; Alshorman & Bataineh, 2024). In the South African context, where digital inequalities and infrastructural disparities persist (Jere et al., 2023; Tarisayi, 2024), examining teacher cognition becomes even more significant. However, rather than approaching this issue solely from a systemic deficit perspective, this study foregrounds the lived experiences of pre-service teachers within a specific ITE programme, recognizing them as active agents negotiating technological change.

The primary significance of this research lies in its contribution to teacher education. By illuminating how AI perceptions and preparedness are constructed during pre-service training, the study offers empirically grounded insights for curriculum developers, teacher educators, and academic institutions seeking to integrate AI meaningfully within Life Sciences methodology courses. Secondary implications may extend to broader educational policy discussions and long-term school-level integration strategies. However, the immediate contribution remains within the sphere of Initial Teacher Education, where foundational technological dispositions are formed.

By centering the voices of Life Sciences pre-service teachers, this study advances scholarly understanding of AI preparedness within an African ITE context and contributes to the growing international discourse on teacher cognition, disciplinary specificity, and emerging educational technologies.

1.6 Research Objectives and Questions

In response to the problem outlined above, this study seeks to develop a nuanced understanding of AI integration within Initial Teacher Education by foregrounding pre-service teachers' current preparedness, perceptions, and contextual experiences. The focus is not merely on speculative future practice, but on how Life Sciences pre-service teachers presently experience and interpret AI within their teacher preparation programme, and how these experiences shape their preparedness to integrate AI pedagogically.

The study is guided by the following objectives:

- To investigate the preparedness of Life Sciences pre-service teachers regarding AI integration in Life Sciences teaching.
- To explore pre-service teachers' perceptions and beliefs about the role and value of AI in Life Sciences education.
- To examine how experiences within the ITE programme and broader institutional context shape pre-service teachers' preparedness to engage with AI.

Accordingly, the study seeks to answer the following research questions:

1. What levels and forms of preparedness do Life Sciences pre-service teachers report regarding AI integration in subject teaching?
2. What are pre-service teachers' perceptions and beliefs about the role and potential of AI in Life Sciences education?
3. How do experiences within the ITE programme and broader institutional context shape pre-service teachers' preparedness to integrate AI?

Together, these questions allow for an integrated exploration of preparedness, perception, and context within the lived experiences of pre-service teachers.

1.7 Definition of Concepts and Relationship of Key Concepts

To ensure conceptual clarity, this study distinguishes between Artificial Intelligence, experiences, beliefs, perceptions, self-efficacy, readiness and preparedness. Although these concepts are related, they refer to different aspects of pre-service teachers' engagement with AI.

1.7.1 Artificial Intelligence (AI)

Artificial Intelligence is broadly understood as computational technology capable of performing tasks that typically require human intelligence, such as learning, reasoning, problem-solving, language processing, and decision-making. Within education, AI has evolved into a rapidly expanding field that applies these intelligent technologies to support teaching, learning, assessment, and educational decision-making. Guo, Zheng, and Zhai (2024) note that AI in education encompasses a wide range of applications, including intelligent tutoring systems, automated grading, recommender systems, learning analytics, educational robots, and generative AI, reflecting the increasing integration of AI across educational contexts (Guo, Zheng, & Zhai, 2024). For the purposes of this study, Artificial Intelligence refers specifically

to AI technologies that have the potential to support teaching and learning within secondary school Life Sciences classrooms.

1.7.2 Artificial Intelligence as a Pedagogical Tool

Artificial Intelligence may serve a variety of educational purposes; however, its value in education lies not merely in the technology itself but in how it is used to facilitate teaching and learning. Ramnarain et al. (2025) argue that preparing pre-service science teachers for AI integration requires moving beyond technological awareness towards the pedagogically meaningful use of AI within classroom practice. Similarly, the growing body of AI in education research demonstrates that AI is increasingly being used to support instructional activities such as personalized learning, adaptive feedback, intelligent tutoring, assessment, and learner engagement rather than functioning solely as a technological innovation (Ramnarain et al., 2025; Guo, Zheng, & Zhai, 2024). In this study, AI as a pedagogical tool refers to the purposeful use of AI to directly support teaching and learning in Life Sciences classrooms. This definition excludes administrative or professional uses of AI, such as drafting emails or organizing documents, where these activities do not directly facilitate learner engagement or classroom instruction.

1.7.3 AI Tools

While Artificial Intelligence refers to the underlying technology, AI tools refer to the software applications through which users interact with AI capabilities. These applications perform specific functions such as generating content, providing explanations, analysing information, offering feedback, supporting assessment, and facilitating learner interaction. Guo, Zheng, and Zhai (2024**)** identify generative AI alongside intelligent tutoring systems, recommender systems, learning analytics, automated grading, and educational robots as major AI applications that have become increasingly prominent within educational research and practice (Guo, Zheng, & Zhai, 2024**).** For the purposes of this study, AI tools refer to AI-powered applications, such as ChatGPT, Microsoft Copilot, Google Gemini, DeepSeek, and Perplexity, that pre-service Life Sciences teachers use or anticipate using to support teaching, learning, assessment, and learner engagement.

1.7.4 Educational Technology

Educational technology is broader than Artificial Intelligence and refers to the ethical study and application of theory, research, and practice to improve learning and performance through the design, development, implementation, management, and evaluation of learning processes and resources. The Association for Educational Communications and Technology (AECT) defines educational technology as a discipline concerned with advancing knowledge, improving learning, and empowering learners through the strategic use of appropriate processes and resources (AECT, 2023).

Artificial Intelligence therefore represents one category of educational technology, alongside other technologies such as learning management systems, multimedia applications, virtual learning environments, simulations, and educational software. Although all AI applications used for teaching are educational technologies, not all educational technologies employ AI. For the purposes of this study, educational technology refers to the broad range of digital technologies used to facilitate teaching and learning, with particular emphasis on AI as one emerging educational technology that supports Life Sciences teaching.

1.7.5 Digital Technologies

Digital technologies refer to electronic technologies that create, process, store, communicate, or manage digital information. Within education, digital technologies include computers, tablets, smartphones, interactive whiteboards, internet-based platforms, educational software, and online communication tools. UNESCO recognises digital technologies as encompassing a wide range of technologies that support teaching, learning, communication, and educational management, with Artificial Intelligence representing one component of this broader digital ecosystem (UNESCO, 2025).

Accordingly, digital technologies constitute the broadest category of technology discussed in this study. Educational technology represents the application of these technologies for educational purposes, while Artificial Intelligence refers to a specialised branch of digital technology capable of simulating aspects of human intelligence. AI tools are therefore specific applications through which users interact with AI technologies. For the purposes of this study,

digital technologies refer to the broader technological environment within which educational technologies and AI applications operate.

1.7.6 Readiness

Readiness refers to the technological, pedagogical, content-specific and contextual knowledge that pre-service teachers possess, or report possessing, to evaluate and integrate AI meaningfully into Life Sciences teaching. It extends beyond general awareness, personal familiarity or willingness to use AI. It includes knowledge of AI tools, an understanding of how they can support Life Sciences content and pedagogy, and awareness of ethical, institutional and contextual considerations affecting their use (Guo et al., 2024; Deb & Sofal, 2026; Ramnarain et al., 2025).

1.7.7 Experiences

Experiences refer to the formal and informal encounters through which individuals interact with and develop an understanding of a particular phenomenon. Within teacher education, these experiences may arise through coursework, teaching practice, professional learning opportunities, or personal engagement with educational technologies. Recent studies on AI in teacher education suggest that pre-service teachers' experiences with AI are shaped largely by independent exploration, academic use, and exposure during teacher education rather than through formal AI instruction, and that these experiences influence how they subsequently engage with AI in educational contexts (Arek-Bawa & Reddy, 2025; Ramnarain et al., 2025). For the purposes of this study, experiences refer to pre-service Life Sciences teachers' formal and informal encounters with AI through personal use, academic activities, teaching practice, interactions with lecturers and peers, and other educational contexts.

1.7.8 Beliefs

Beliefs are psychologically held understandings, propositions, or assumptions that individuals accept as true and that influence how they interpret experiences and make decisions (Pajares, 1992). Unlike factual knowledge, beliefs are developed over time through personal experiences and are often resistant to change. Ertmer (2005) further argues that teachers' pedagogical beliefs are among the most influential factors shaping whether and how they integrate technology into their classroom practice, even when technological resources and training are

available. Teachers' beliefs therefore influence how they evaluate the educational value of emerging technologies and the role these technologies should play in teaching and learning. In this study, beliefs refer to pre-service teachers' assumptions and convictions about the educational role of AI, including its potential benefits, limitations, ethical implications, and appropriateness for Life Sciences teaching.

1.7.9 Perceptions

Perceptions refer to individuals' interpretations and evaluations of a phenomenon based on their prior experiences and beliefs. Within the Technology Acceptance Model, teachers' perceptions of a technology are reflected through their evaluations of its usefulness and ease of use, which influence their willingness to adopt that technology (Davis, 1989). In the context of AI in education, Guo, Shi, and Zhai (2025) conceptualise teachers' perceptions through constructs such as perceived usefulness, perceived ease of use, behavioural intention, and self-efficacy, recognising these as important dimensions influencing teachers' acceptance of AI.

Within this study, experiences, beliefs, perceptions and preparedness are understood as related but distinct concepts. Pre-service teachers' formal and informal experiences with AI contribute to the development of beliefs about its educational role, which shape their perceptions of its usefulness, ease of use and relevance to Life Sciences teaching. These perceptions may influence their confidence and willingness to engage with AI, but preparedness refers more specifically to the technological, pedagogical, content-specific and contextual knowledge required for meaningful integration.

1.7.10 Perceived Usefulness (PU)

Perceived Usefulness is defined by Davis (1989) as the degree to which an individual believes that using a particular technology will enhance his or her job performance. Within educational contexts, perceived usefulness reflects teachers' beliefs that a technology can improve teaching effectiveness, learner engagement, instructional quality, or learning outcomes. Building on Davis' conceptualization, Guo, Shi, and Zhai (2025) operationalize perceived usefulness in the context of Artificial Intelligence as teachers' beliefs that AI can improve teaching efficiency, support instructional decision-making, enhance student learning, and contribute positively to classroom practice. For the purposes of this study, perceived usefulness refers to pre-service Life Sciences teachers' beliefs that AI can enhance teaching and learning by improving

instructional effectiveness, supporting learner engagement, facilitating conceptual understanding, and enriching classroom practice.

1.7.11 Perceived Ease of Use (PEOU)

Davis (1989) defines Perceived Ease of Use as the degree to which an individual believes that using a particular technology will be free of effort. Within teacher education, this construct reflects teachers' perceptions of how easy AI technologies are to learn, understand, and incorporate into teaching practice. Guo, Shi, and Zhai (2025) similarly conceptualise perceived ease of use as teachers' confidence that AI technologies can be used without excessive complexity or difficulty, thereby influencing their acceptance of AI in educational settings. For the purposes of this study, perceived ease of use refers to pre-service Life Sciences teachers' perceptions of how easy AI tools are to learn, use, and integrate into Life Sciences teaching.

1.7.12 Behavioural Intention

Within the Technology Acceptance Model, Davis (1989) conceptualizes behavioral intention as an individual's intention or willingness to use a particular technology, which serves as the immediate predictor of actual technology use. In the context of AI in education, Guo, Shi, and Zhai (2025) describe behavioral intention as teachers' intention to adopt and use AI technologies within their teaching practice, recognising it as a key indicator of AI acceptance. For the purposes of this study, behavioral intention refers to pre-service Life Sciences teachers' expressed intention or willingness to integrate AI into their future classroom practice. Although behavioral intention reflects participants' motivation to use AI, it does not necessarily indicate that they possess the pedagogical capability required for effective AI integration.

1.7.13 Self-Efficacy

Bandura (1997) defines self-efficacy as individuals' beliefs in their capability to organize and execute the actions required to achieve specific performance outcomes. Within educational technology, self-efficacy reflects teachers' confidence in their ability to use technology effectively to support teaching and learning. Drawing on Bandura's work, Guo, Shi, and Zhai (2025) conceptualise AI self-efficacy as teachers' confidence in their ability to use AI technologies effectively within educational contexts. For the purposes of this study, self-

efficacy refers to pre-service Life Sciences teachers' confidence in their ability to use AI meaningfully to support teaching and learning. Self-efficacy is recognised as distinct from preparedness because teachers may feel confident in using AI while still lacking the pedagogical knowledge required for its effective classroom integration.

1.7.14 Preparedness for AI Integration

The concepts of preparedness and readiness are frequently used interchangeably within the educational technology and Artificial Intelligence literature to describe teachers' capacity to integrate emerging technologies into their professional practice. However, there is no universally accepted definition of either construct, with studies conceptualising teacher preparedness in terms of knowledge, skills, attitudes, confidence, motivation, technological competence, pedagogical capability, and contextual awareness. For example, Wang et al. (2023) define teachers' AI preparedness as a state of preparedness characterised by teachers' cognition, ability, vision, and ethical considerations regarding the educational use of AI. Similarly, Guan et al. (2025) conceptualise pre-service teachers' preparedness for AI-integrated education as encompassing teachers' perceptions, capabilities, and evolving professional identities, while recent studies on science teacher education emphasise that preparedness for AI integration extends beyond technological familiarity to include pedagogical capability and meaningful classroom application (Ramnarain et al., 2025).

Drawing on these perspectives, this study adopts the term preparedness rather than readiness because it more appropriately reflects the study's focus on pre-service teachers' perceived capability to integrate AI into future classroom practice. Preparedness is understood as a multidimensional construct informed by teachers' experiences with AI, their beliefs and perceptions regarding its educational value, their confidence in using AI, their behavioural intentions towards AI adoption, and their perceived pedagogical capability to integrate AI meaningfully within Life Sciences teaching. This conceptualisation is informed by the Technology Acceptance Model (Davis, 1989), which explains the motivational dimensions of AI adoption, and the adapted Artificial Intelligence Pedagogical Knowledge (AIPK) framework, which explains the pedagogical knowledge required for effective AI integration (Mishra & Koehler, 2006; Deb & Sofal, 2026).

For the purposes of this study, preparedness for AI integration refers to pre-service Life Sciences teachers' perceived motivational, pedagogical, and contextual capability to integrate

AI meaningfully into teaching and learning. Preparedness therefore extends beyond awareness, willingness, confidence, or technological competence alone. Instead, it represents an overarching construct encompassing teachers' perceptions of AI, their behavioural intentions towards AI adoption, and their perceived AI-related.

1.8 Chapter Summary

This chapter introduced the study by situating it within the expanding global discourse on Artificial Intelligence in education, with particular emphasis on its implications for Initial Teacher Education and Life Sciences teaching. It highlighted the growing recognition that the successful integration of emerging technologies depends not only on access to tools, but on teachers' perceptions, beliefs, and knowledge structures. Within this context, the chapter foregrounded the importance of examining how pre-service teachers begin to understand and engage with Artificial Intelligence during their professional preparation.

The chapter further established the research problem by identifying a significant gap in the literature, namely the limited attention given to Artificial Intelligence-specific preparedness among pre-service teachers, particularly within the South African context and within subject-specific domains such as Life Sciences. It emphasised that while global research on technology integration is expanding, much of it remains focused on in-service teachers or general digital tools, thereby overlooking the formative role of Initial Teacher Education in shaping future teaching practice. The purpose of the study was therefore clearly articulated, alongside its significance in contributing to both local and international scholarship.

Finally, the chapter outlined the research objectives and questions guiding the study, establishing a focused inquiry into pre-service teachers' preparedness, perceptions, and contextual experiences of Artificial Intelligence integration. By doing so, the chapter provided a clear conceptual and contextual foundation for the study, positioning it within ongoing discussions on teacher cognition, technological change, and disciplinary pedagogy. This foundation informs the literature review presented in the following chapter.

CHAPTER TWO

LITERATURE REVIEW AND THEORETICAL FRAMEWORK

2.1 Introduction

This chapter examines the body of scholarship relevant to AI integration in education and establishes the theoretical framework underpinning this study. It begins by reviewing international and local literature on AI in teaching and learning, with particular attention to pre-service teacher preparedness, subject-specific pedagogy, and the emerging role of generative AI in educational contexts. The review identifies key debates, opportunities, and concerns surrounding AI adoption in classrooms, while highlighting the limited research focusing specifically on Life Sciences pre-service teachers within the South African context.

Building on this review, the chapter then outlines the theoretical framework guiding the study. Drawing on technology integration and acceptance theories, the framework provides conceptual tools for analysing how pre-service teachers perceive AI, how they evaluate its usefulness and feasibility, and how contextual realities influence their preparedness for implementation. By integrating empirical literature with theoretical positioning, this chapter establishes the conceptual foundation upon which the study's research design, analysis, and interpretation are constructed.

2.2 Artificial Intelligence as a Pedagogical Tool in Education

Artificial intelligence has become increasingly integrated into education, shifting from being viewed primarily as a technological innovation to being recognised for its potential to enhance teaching and learning (Holmes & Tuomi, 2022; Zhai, 2025). While early applications of AI in education focused on intelligent tutoring systems and adaptive learning environments, recent developments in generative AI have expanded its educational applications to include personalized learning, automated feedback, content generation, collaborative learning, and instructional support (Luckin et al., 2016; Kasneci et al. 2023). Consequently, contemporary educational research increasingly conceptualizes AI according to its pedagogical contribution rather than its technological sophistication, emphasizing how AI can support meaningful learning experiences and improve educational outcomes (Zhai, 2025).

Within educational contexts, AI is best understood as a pedagogical tool when it directly facilitates teaching and learning rather than merely performing technological functions. Pedagogical tools are resources intentionally used to promote learners' understanding, engagement, critical thinking, collaboration, assessment, and knowledge construction. Accordingly, AI functions as a pedagogical tool when it supports activities such as adaptive tutoring, personalized instruction, intelligent feedback, inquiry-based learning, formative assessment, and collaborative problem-solving (Luckin et al., 2016; Deb & Sofal, 2026). As Abu-Gweder (2026) argues, the educational significance of AI lies not in the technology itself but in teachers' ability to integrate it meaningfully into pedagogical practice to improve learning. Similarly, Holmes and Tuomi (2022) contend that AI should complement teachers' professional judgement rather than replace it, enabling teachers to create more responsive, learner-centered instructional environments.

However, the pedagogical role of AI should be distinguished from its use as a tool for teacher productivity and instructional preparation. The emergence of generative AI applications such as ChatGPT has enabled teachers to generate lesson plans, prepare teaching materials, design assessments, summarize academic content, and develop classroom resources more efficiently (Zhai, 2024; Nyaaba et al., 2024). Although these applications contribute significantly to teachers' professional practice, they primarily support instructional planning rather than pedagogy itself because learners are not directly interacting with the AI during the teaching and learning process. By contrast, AI assumes a pedagogical role when learners engage directly with intelligent systems that personalize instruction, provide adaptive feedback, scaffold conceptual understanding, or facilitate inquiry-based learning experiences. Distinguishing between teacher-facing AI that supports instructional preparation and learner-facing AI that directly facilitates learning provides greater conceptual clarity regarding the educational role of AI and aligns with recent discussions on the pedagogical integration of generative AI (Wang et al., 2025)

A further conceptual distinction is required between AI, AI tools, educational technology, and digital technologies. Digital technologies constitute the broadest category and include electronic devices, software, internet-based platforms, and communication technologies used across different sectors of society. Educational technology represents a subset of these technologies that are specifically designed or adapted to facilitate teaching and learning,

including learning management systems, multimedia resources, educational software, simulations, virtual reality (VR), and online collaboration platforms (Holmes & Tuomi, 2022). Artificial intelligence, however, is not synonymous with educational technology. Rather, AI refers to computational systems capable of performing tasks associated with human intelligence, such as recognizing patterns, learning from data, generating content, adapting responses, and supporting decision-making (Luckin et al., 2016). AI tools therefore represent educational technologies that incorporate these intelligent capabilities to support teaching and learning.

Similarly, simulations and virtual reality should not automatically be regarded as AI technologies. While recent developments have enabled AI to be embedded within simulation and VR environments through adaptive learner modelling, intelligent feedback, conversational agents, and personalized learning experiences, simulations and VR can also operate independently of artificial intelligence. Consequently, AI-powered simulations and AI-enhanced virtual reality environments should be distinguished from simulations and virtual reality as educational technologies in their own right. Maintaining these conceptual distinctions is particularly important because the increasing convergence of digital technologies may lead educators to associate all advanced educational technologies with AI, thereby obscuring the unique pedagogical affordances that artificial intelligence contributes to teaching and learning.

2.3 Pre-Service Teachers' Perceptions of Artificial Intelligence in Education

AI continues to shape education globally, introducing tools and systems that support personalized learning, automate routine tasks, and enhance pedagogical effectiveness. Preservice teachers often engage with AI from a prospective and interpretive standpoint, forming early views based on their training, experiences, and personal digital fluency.

These perceptions reflect how pre-service teachers interpret AI's potential role in the classroom, what it might enable or limit, and how it aligns with their evolving teaching identity. As Mnguni (2024) argues that pre-service teachers are at a "seminal point in their careers," where they are forming the technological pedagogical content knowledge that will underpin their teaching philosophies. Unlike in-service teachers, whose perceptions are often shaped by classroom realities, pre-service teachers may view AI with a mix of enthusiasm, curiosity, and uncertainty (Caner & Aydin, 2021; Mnguni, 2023).

Studies highlight that many pre-service teachers view AI as an opportunity to enhance teaching and learning. In Mnguni's (2023) study, participants identified pedagogical benefits such as personalized learning, improved interactivity, and the ability to make complex content accessible, especially in Life Sciences Grade 12 topics like genetics and homeostasis. Similarly, Akkaya (2016) found that pre-service teachers recognized AI's potential to reduce instructional load, personalize learning, and improve feedback efficiency. These perceptions are often linked to higher levels of digital confidence and openness to innovation, which are fostered through intentional training experiences.

However, these perceptions are not uniformly optimistic. Several studies (Caner & Aydin, 2021; Pokrivčáková, 2023; Karataş & Yüce, 2024) report that pre-service teachers often express cautious scepticism, particularly where AI is seen as too complex, inaccurate in some instances, impersonal, or inaccessible. Mnguni (2024) reports apprehension about the digital divide, fears of over-reliance on technology, and worries about undermining human teacher roles. This is echoed in broader research, where AI is seen as a "black box"; lacking transparency in how it makes decisions or recommends actions (Ng et al., 2021). These critical perceptions are often shaped by a lack of exposure to AI tools during training, or by practicum experiences in under-resourced schools where such tools are absent.

Perceptions are also influenced by peer discussions, lecturer emphasis, and the visibility of AI in policy and curriculum conversations. As Aslan & Zhu (2016) point out, teacher education courses that include AI-infused content foster more informed and open perceptions. Conversely, where exposure is low, AI is often perceived as intimidating or irrelevant. Furthermore, studies (Shilenge & Ramaila, 2020; Caner & Aydin, 2021; Yue et al., 2024) show that perceptions are shaped by school environments during teaching practice. For instance, pre-service teachers placed in under-resourced schools become more aware of practical challenges and may internalize institutional scepticism toward technology (Venketsamy & Hu, 2022; Mnguni, 2024).

In the context of Life Sciences, the literature consistently affirms that pre-service teachers' perceptions of AI are complex and context dependent, however, become especially important. As Mnguni (2024) argues, these perceptions are not merely passive opinions, they represent early indicators of how AI might be adopted in South African classrooms in the near future.

Life Sciences often requires learners to visualise unseen or dynamic processes, such as genetic expression or ecological systems, which can be difficult to teach using traditional methods. When pre-service teachers perceive AI as capable of enhancing these explanations, they are more likely to imagine themselves using it in future classrooms. Thus, perceptions directly inform their anticipated practices, particularly in how they foresee AI supporting conceptual understanding and learner-centred pedagogy (Mnguni, 2023).

2.4 Anticipated Use of AI in Life Sciences Classrooms

As the influence of AI continues to expand in the education sector, there is growing interest in understanding how teachers, not just those in service but also those in training, envision using these tools in their future practice. In Life Sciences education, AI holds promise for enhancing the teaching of complex biological concepts, facilitating personalized learning, and supporting formative assessment (Mnguni, 2023). However, how pre-service teachers anticipate using AI remains a largely under-explored area of research, particularly within the South African context.

Mnguni (2024) found that pre-service Life Sciences teachers expressed keen interest in using AI to make content more interactive and accessible, enhancing the teaching of complex, abstract biological concepts. Participants anticipated that AI-powered simulations could help illustrate abstract scientific processes such as DNA replication, osmosis, homeostasis and population dynamics, topics often perceived as challenging due to their microscopic or systemic nature. By interacting with dynamic visualisations or adaptive tutorials, learners could gain a more intuitive grasp of these phenomena. These anticipated applications reflect a belief that AI can mediate understanding where textbook-based or static instruction falls short.

Beyond content delivery, pre-service teachers imagined AI to assist with automating routine teaching tasks, learner assessment, and differentiated instruction, thereby freeing up time for more learner-focused teaching strategies. Others anticipated that AI tools could assist with administrative tasks, such as generating reports, tracking learner progress, or managing large volumes of assessment data (Akkaya, 2016; Yue et al., 2024).

This optimism, however, coexists with cautious pragmatism. Pre-service teachers acknowledged that their envisioned use of AI would depend on systemic support, including availability of infrastructure, training support, and their own confidence with technology. As

highlighted by Yue et al. (2024), the anticipation of using AI is often conditional, shaped by access to appropriate tools, teacher autonomy, and confidence in their own technological abilities. This reflects the close relationship between perceptions and practice, as explored in Section 3.1: when pre-service teachers perceive AI as useful and manageable, they are more likely to anticipate incorporating it into their classrooms.

In some cases, anticipated use was framed not just as a pedagogical strategy, but as a means of inclusion. Participants in Mnguni's (2024) study imagined using AI to provide multilingual explanations or to scaffold learning for struggling students. Such uses reflect a growing awareness of AI's potential to support diverse learner needs, a particularly relevant consideration in South African public schools.

Furthermore, some respondents in Mnguni's study raised ethical and philosophical concerns about over-reliance on AI in teaching, particularly in contexts where relational and value-driven education is prioritized. There was a shared recognition that while AI may enrich instruction, it should not replace the pedagogical agency and emotional intelligence of human teachers.

This concern highlights the ethical dimension of AI use; pre-service teachers want to ensure that technology enhances rather than replaces relational and inquiry-based teaching practices. These reflections further demonstrate that anticipated AI use is not simply a matter of technical optimism, but a nuanced consideration of pedagogy, context, and identity.

2.5 Contextual and Psychological Factors Influencing Preparedness

Pre-service teachers' preparedness to integrate AI in their future classrooms is influenced by a combination of cognitive, institutional, and contextual factors (Yue, Wang, & Qiu, 2024). In this study, this subsection focuses on the contextual and psychological influences on preparedness, including confidence, training, support structures, and systemic barriers. A deeper discussion of preparedness is provided in Section 3.5.

These include their attitudes and beliefs, self-efficacy, institutional support, prior training, and exposure to AI tools (; Howard, Tondeur, & Ma, 2022). Understanding these factors is vital

for designing effective teacher education programs that prepare future Life Sciences educators for AI-enhanced teaching.

One of the most prominent influences on preparedness is self-efficacy; the belief in one's own ability to perform a task (Bandura, 1997). Pre-service teachers who perceive themselves as competent in using AI tools are more likely to express preapredness and intention to implement them in future classrooms (Caner & Aydin, 2021). In Mnguni's (2024) study, self-efficacy emerged through participants' confidence levels in exploring or applying AI to support Life Sciences teaching. Participants who had prior exposure to digital technologies during training or teaching practice felt more prepared to engage with AI, while those who lacked this exposure often doubted their abilities.

Closely related to self-efficacy is the issue of teacher training. Several studies have noted that South African teacher education programmes often prioritise general ICT skills over AI-specific competencies (Venketsamy & Hu, 2022; Shilenge & Ramaila, 2020). While most pre-service teachers are introduced to digital pedagogy, few are provided with structured opportunities to explore AI tools or apply them in subject-specific contexts. Mnguni (2024) and Aslan and Zhu (2016) both highlight the importance of hands-on training, exposure to real-life classroom applications, and alignment between university coursework and school-based experience. Without these components, preparedness remains low, even among digitally literate students.

Institutional constraints, such as limited curriculum time, outdated programme content, and lack of qualified staff to teach AI concepts, also contribute to low levels of preparedness. Teacher education programmes may not currently integrate AI into teaching methodology modules or practicum supervision. These gaps represent not just individual barriers but broader systemic obstacles to preparing future teachers for AI-enhanced education.

Another influence on preparedness is access to infrastructure. Pre-service teachers placed in under-resourced schools during their practicum often reported difficulties imagining AI use in practice, due to unreliable internet, lack of devices, or absence of technical support (Shilenge & Ramaila, 2020; Nkula & Krauss, 2014). Even where teacher candidates perceive AI as useful, a lack of enabling conditions can significantly reduce their sense of preparedness.

Subjective norms: what peers, lecturers, and social expectations, including from mentors or experts, also shape preparedness. The Theory of Planned Behavior (Ajzen, 2014), which underpins Mnguni's work, highlights how perceptions of what others, such as peers, mentors, and institutional authorities, expect can influence behavioral intentions. For example, if pre-service teachers perceive AI use as supported or encouraged by influential figures in their professional environment, they are more likely to embrace its integration. Mnguni (2023) found that peer influence, parental attitudes, and school governing body policies all played a role in shaping participants perceived preparedness and decision-making regarding AI adoption.

Lastly, AI-specific literacy; understood as both technical knowledge and ethical understanding of AI systems, is emerging as a vital determinant of preparedness. Howard et al. (2022) argue that general digital literacy is no longer sufficient; teachers must develop AI-specific competencies, including understanding algorithmic bias, data privacy, and pedagogical alignment of AI tools. Without these competencies, even confident pre-service teachers may be unprepared to critically assess and meaningfully apply AI in Life Sciences classrooms.

2.6 Challenges and Concerns about AI Integration in Life Sciences Classrooms

While several studies (e.g., Mnguni, 2023; Mnguni, 2024; Ayanwale et al., 2022; Caner & Aydin, 2021; AlKanaan, 2022; Yue et al., 2024) have highlighted challenges that influence both perceptions and preparedness pre-service teachers face in integrating AI into teaching. Much of this literature tends to generalize across contexts and subject areas. Relatively little is known about how these challenges are experienced specifically within Life Sciences education or among South African pre-service teachers. This section synthesizes key challenges reported in existing research, which this study seeks to explore in greater depth within a localized and subject-specific context.

One widely cited challenge is limited exposure and training. Many pre-service teachers enter their training programs with little to no understanding of what AI is, let alone how it might be used in classroom settings. As Mnguni (2023; 2024) notes, participants expressed a lack of hands-on experience with AI tools and uncertainty about how to apply them to teaching complex Life Sciences topics such as genetics or cellular processes. This aligns with findings

from Yue et al. (2024), who argue that pre-service teachers' preparedness is often hindered by the absence of AI-specific modules in teacher education programs.
This lack of exposure reduces self-efficacy and, in turn, lowers their sense of preparedness to implement AI in their future classrooms. Caner and Aydin (2021) found that pre-service teachers often lack confidence in their ability to integrate AI effectively due to their unfamiliarity with the technology and fear of making pedagogical errors. Similarly, Mnguni (2023) reported that many South African pre-service teachers viewed AI as overly technical and intimidating, especially in contexts where their teacher training had not equipped them with AI-related knowledge or pedagogical strategies.

Most South African pre-service programmes focus on general ICT skills and offer few if any structured opportunities to explore AI pedagogy in subject-method modules (Venketsamy & Hu, 2022; Shilenge & Ramaila, 2020). Where AI is addressed, it is often in passing or through theoretical discussions rather than through hands-on experiences. This lack of curricular alignment means that even students who are open to innovation may feel unprepared to use AI practically. Thus, curriculum design itself becomes a structural barrier to preparedness.

Infrastructure and resource constraints represent another significant barrier. In some parts of South Africa, disparities in technological access between urban and rural schools limit both the feasibility of AI use and pre-service teachers' confidence in applying what they learn. Shilenge and Ramaila (2020) highlight the impact of unequal access to internet connectivity, devices, and basic digital infrastructure in teacher education institutions. Pre-service teachers who train in low-resource environments often internalise limitations and adjust their teaching plans accordingly, which may exclude AI altogether. As Nkula and Krauss (2014) argue, preparedness cannot be cultivated in environments where access to basic tools is inconsistent.

Ethical concerns also shape pre-service teachers' perceptions and preparedness. AlKanaan (2022) and Ayanwale et al. (2022) report that future educators are often apprehensive about issues such as data privacy, algorithmic bias, and the potential erosion of teacher–learner relationships. These concerns, echoed in Mnguni (2024), reflect deeper philosophical tensions around the role of AI in human-centred educational contexts, especially in societies that value relational teaching practices and ethical reflection.

While these challenges have been broadly identified in global and South African contexts, few studies have examined how they manifest specifically in Life Sciences education, or how they are interpreted by pre-service teachers nearing the transition into full-time teaching. This study aims to fill that gap by offering a focused, contextualized account of these concerns as experienced by third and fourth-year B.Ed. students at a South African university.

2.7 Dimensions of Preparedness for AI Integration

Preparing teachers to integrate AI into teaching has become an increasingly important focus of teacher education research as AI technologies become more accessible within educational contexts. Although the terms *preparedness* and *readiness* are frequently used interchangeably, they generally refer to a teacher's overall capacity to adopt and integrate AI meaningfully within professional practice. Recent studies suggest that preparedness extends beyond technological familiarity and encompasses the knowledge, perceptions, confidence, pedagogical capability, and contextual awareness required to integrate AI effectively and responsibly in teaching and learning (Guan et al., 2025; Wang et al., 2023; Ramnarain et al., 2025).

Preparedness is therefore best understood as a multidimensional construct rather than a single competency. While some studies emphasise technological competence, others highlight teachers' pedagogical capability, ethical awareness, professional confidence, or institutional support as equally important dimensions of preparedness (Deb & Sofal, 2026; Guan et al., 2025). In science education particularly, preparedness involves not only knowing how AI functions but also understanding when, why, and how AI can be used to enhance disciplinary teaching while supporting meaningful learner engagement and scientific understanding (Ramnarain et al., 2025).

2.7.1 Conceptual Preparedness

Conceptual preparedness refers to teachers' understanding of what Artificial Intelligence is, how it differs from broader digital technologies, and the educational purposes for which AI can appropriately be used. Developing conceptual preparedness requires more than recognizing AI applications or using AI-powered tools. Rather, teachers need to understand AI as a distinct

educational technology with particular capabilities, limitations, and ethical implications (Guan et al., 2025; Guo, Zheng, & Zhai, 2024).

Recent literature suggests that many teachers and pre-service teachers demonstrate familiarity with AI through everyday exposure to generative AI applications and other digital technologies. However, this familiarity does not necessarily indicate conceptual understanding of AI or its educational applications (Karataş & Yüce, 2024; Pokrivčáková, 2023). Consequently, teachers may possess general digital competence while still lacking the AI-specific knowledge required to distinguish AI from broader educational technologies and to make informed pedagogical decisions regarding its use.

Developing conceptual preparedness therefore requires teacher education programmes to move beyond introducing AI tools towards fostering AI literacy, enabling teachers to critically understand the capabilities, limitations, ethical considerations, and appropriate educational applications of AI.

2.7.2 Pedagogical Preparedness

Pedagogical preparedness refers to teachers' capability to integrate AI meaningfully within teaching and learning rather than simply using AI technologies. This involves selecting AI tools that align with curriculum goals, employing appropriate instructional strategies, supporting learner engagement, and ensuring that AI contributes to meaningful learning rather than becoming an end in itself (Deb & Sofal, 2026; Guan et al., 2025).

Within Life Sciences education, pedagogical preparedness includes recognizing how AI can support the teaching of complex biological concepts through personalized explanations, adaptive feedback, visualization of abstract processes, formative assessment, and inquiry-based learning. The emphasis therefore shifts from simply using AI to using AI purposefully to enhance conceptual understanding and scientific reasoning (Ramnarain et al., 2025).

Research consistently suggests that although many pre-service teachers express positive attitudes towards AI, they often report uncertainty regarding how AI should be incorporated into subject-specific teaching. This indicates that pedagogical preparedness develops through structured opportunities to apply AI within authentic teaching contexts rather than through technological exposure alone (Seufert et al., 2021; Deb & Sofal, 2026).

2.7.3 Contextual Preparedness

Preparedness for AI integration is also influenced by the broader educational context within which teachers learn and teach. Contextual preparedness encompasses institutional support, access to appropriate technologies, professional development opportunities, curriculum expectations, ethical guidance, and school infrastructure that collectively shape teachers' opportunities to integrate AI effectively (Ramnarain et al., 2025; Wang et al., 2023).

Within South Africa, contextual preparedness is particularly important because schools and teacher education institutions operate within diverse technological environments characterized by unequal access to digital resources, varying institutional policies, and emerging approaches to AI integration. Consequently, preparedness cannot be understood solely as an individual teacher characteristic but must also be considered in relation to the educational environments that enable or constrain AI integration (Venketsamy & Hu, 2022; Shilenge & Ramaila, 2020).

These contextual influences reinforce the view that preparing future teachers for AI integration requires more than technological training alone. Effective preparedness depends upon the interaction between teachers' conceptual understanding of AI, their pedagogical capability to use AI meaningfully, and the institutional conditions that support responsible AI integration.

Ultimately, knowledge preparedness for AI integration involves more than just familiarity with digital tools; it requires the professional capability to strategically link technological possibilities with discipline-specific educational goals (Guan et al., 2025).

2.7.4 The South African Context and Research Gaps

Globally, AI is increasingly viewed as a catalyst for innovation in science education, yet in South Africa, the integration of AI into classroom teaching remains in its infancy. The Department of Basic Education has emphasized the need to develop teachers' digital competencies to prepare learners for the Fourth Industrial Revolution (4IR) (DBE, 2019). However, efforts to embed AI into educational systems remain uneven and underdeveloped, particularly within subject-specific domains such as Life Sciences. Current teacher education programmes often prioritise foundational digital skills such PowerPoint and online platforms, without equipping future teachers to engage with intelligent systems such as AI-powered

virtual simulations, AI tutors, or real-time data tools (Shilenge & Ramaila, 2020; Venketsamy & Hu, 2022).

While research on digital learning in STEM subjects is growing in South Africa (Mpungose, 2020; Mahlaba, 2020), there is a tendency to treat 'science' as a broad category, without exploring the unique demands of Life Sciences. For examples, the teaching of abstract concepts like evolution and cellular mechanisms. AI could be a viable tool for effectively simplifying these concepts through adaptive technologies, yet little is known about whether pre-service teachers perceive this tool as useful.

One of the key contextual factors shaping AI integration in South African education is the evolving institutional and policy environment surrounding the use of generative AI. While the Department of Basic Education (2019) has emphasised the development of digital competencies to prepare learners for the Fourth Industrial Revolution (4IR), explicit guidance on the pedagogical integration of AI within teacher education remains limited. More recently, South African higher education institutions have begun developing institutional policies and guidelines governing the responsible use of generative AI. However, policy implementation remains uneven across institutions, with many universities relying on guidelines rather than comprehensive institution-wide AI policies (Chaka et al., 2024). Through an environmental review of 26 South African higher education institutions, Chaka et al. (2024) found that only a small number had publicly available institution-wide AI policy documents, while most institutions were still developing guidance focused primarily on responsible AI use, academic integrity, and ethical considerations. Similarly, Jacobs-Basadien and Pather (2026) argue that although South African universities have increasingly recognised the need for AI governance, institutional responses continue to vary considerably, reflecting the rapidly evolving nature of generative AI adoption within higher education.

Beyond policy development, institutional realities within South African schools also influence the extent to which AI can be meaningfully integrated into classroom practice. Although AI-powered educational technologies offer opportunities to enhance teaching and learning, implementation remains constrained by contextual factors such as unequal access to digital infrastructure, inconsistent internet connectivity, limited technological resources, and school-level regulations governing the use of digital devices. These contextual challenges are

particularly evident in under-resourced schools, where disparities in infrastructure continue to influence teachers' opportunities to implement technology-enhanced pedagogies (Nkula & Krauss, 2014; Shilenge & Ramaila, 2020). Consequently, teachers' preparedness to integrate AI should not be understood solely as an individual competency but also as a function of the institutional environments in which they are expected to teach. As Jacobs-Basadien and Pather (2026) observe, effective AI integration requires institutional support structures, clear governance frameworks, and equitable access to digital resources alongside teachers' pedagogical and technological capabilities.

Another emerging institutional challenge concerns academic integrity and the validity of assessment in the era of generative AI. The widespread availability of tools such as ChatGPT has prompted universities to reconsider existing approaches to assessment, authorship, plagiarism, and responsible AI use. Chaka et al. (2024) identified academic integrity and the responsible use of AI as central themes across South African university AI guidelines, reflecting growing institutional concern regarding the ethical implications of AI-assisted learning. Likewise, Jacobs-Basadien and Pather (2026) report that contemporary institutional policies increasingly emphasize transparency, accountability, disclosure of AI use, and ethical governance as essential principles for responsible AI adoption within higher education. International evidence similarly suggests that universities are moving away from prohibition towards developing assessment approaches that promote ethical AI use while preserving assessment validity and authentic learning (Wang et al., 2024). Collectively, these developments demonstrate that successful AI integration depends not only on teachers' preparedness but also on institutional policies, ethical governance, and assessment practices that support the responsible educational use of AI.

There is also limited research on pre-service teachers' preparedness and anticipated use of ai in subject teaching. Recent studies (e.g., Shilenge & Ramaila, 2020; Mnguni 2023; Mnguni 2024) have begun to address this gap by exploring pre-service teachers' behavioral intentions toward AI integration specifically in Life Sciences education. These studies offer valuable insights into how teacher candidates interpret AI's pedagogical potential, ethical considerations, and institutional constraints. However, as these studies note, behavioral intention does not equate to practical preparedness, nor does it necessarily capture the complex

interplay of contextual realities, personal beliefs, and institutional support mechanisms that influence eventual classroom practices.

Furthermore, studies that focus on AI often examine in-service teachers or look at technology integration through the lens of policy and infrastructure. There is a clear gap in understanding the pedagogical beliefs, exposure, and practical expectations of third and fourth-year pre-service teachers, particularly in Life Sciences, where the potential for AI to transform teaching and learning is considerable, but still under-theorized.

This study seeks to fill that gap by focusing on pre-service Life Sciences teachers at a South African university. It contributes to the existing literature by:

- Exploring not only intentions but also perceptions and anticipated practices.
- Contextualizing these within the specific realities of Life Sciences as a subject.
- Offering qualitative insights grounded in South African socio-educational realities, including school quintile disparities, evolving institutional AI policies, curriculum demands, assessment integrity, and institutional preparedness.
- Informing future curriculum design and teacher training interventions that address both the pedagogical and ethical dimensions of AI in science education.

By situating this study within these intersecting gaps, it not only advances scholarly understanding but also aligns with national educational goals to modernize teacher education and equip future educators with competencies relevant to the digital age.

2.8 Theoretical framework

This study is guided by two complementary theoretical frameworks: the TAM and the TPACK framework. Together, these frameworks provide a powerful lens through which to examine how pre-service Life Sciences teachers perceive, anticipate, and prepare to integrate AI into their future teaching practice. TAM addresses the cognitive and psychological dimensions of acceptance (Davis, 1989), while TPACK provides a pedagogical model for understanding how AI might fit within subject-specific teaching (Mishra & Koehler, 2006).

TAM alone, however, pays limited attention to the pedagogical and content-specific dimensions of technology integration, as well as to the structural realities of South African schooling. Conversely, TPACK foregrounds the interplay between technological, pedagogical

and content knowledge, but does not explicitly account for teachers' perceptions of usefulness or ease of use. In this study, TAM and TPACK are therefore combined into an integrated framework that links context, perceptions of AI, and Life Sciences-specific technological and pedagogical knowledge. The integration of these frameworks enables AI preparedness to be conceptualized as the interaction between technology acceptance, pedagogical capability, and the contextual conditions that shape AI integration within South African Life Sciences education. The relationship between these elements is summarized in Figure 1.

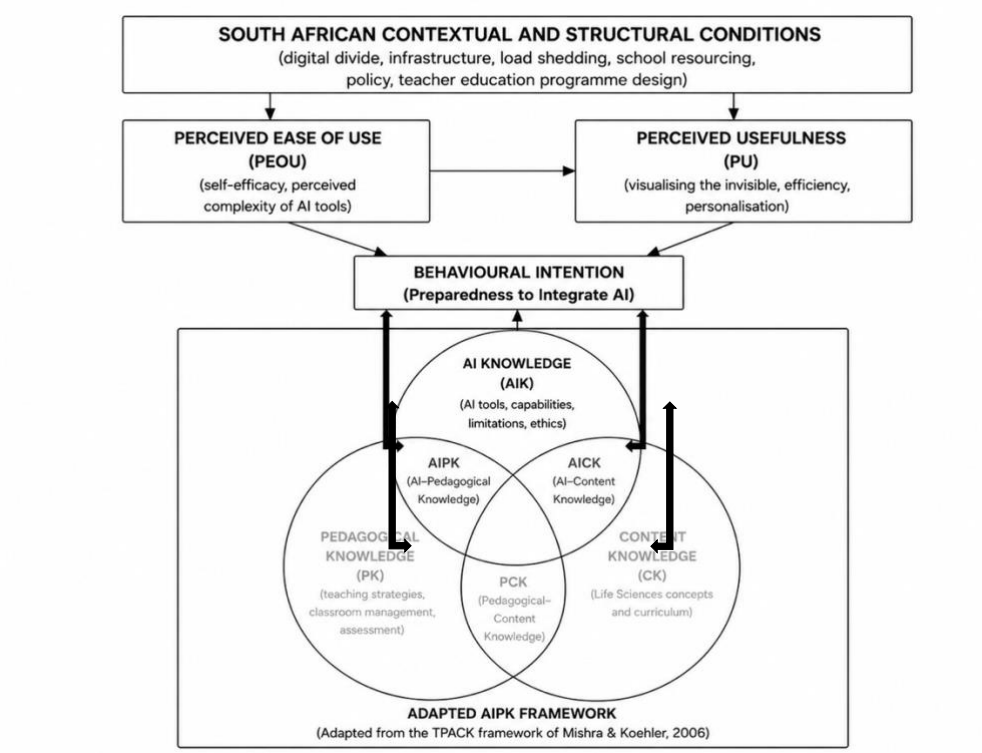


Figure 1: Integrated TAM-TPACK framework for analysing pre-service life sciences teachers' AI preparedness

2.8.1 Technology Acceptance Model

This study draws on the TAM developed by Davis (1989) and refined by subsequent researchers such as Ma and Liu (2005) and Guo, Shi and Zhai (2024). As a foundational theoretical lens for examining how Life Sciences pre-service teachers perceive the integration of AI into their future classrooms. TAM is widely used in educational technology research to explain how users come to accept and adopt new technologies. It posits two primary factors: Perceived Usefulness (PU); the degree to which an individual believes that using a particular technology will improve their job performance, and Perceived Ease of Use (PEOU); the degree to which a person believes that using the technology will be free of effort (Davis, 1989). In the context of education, PU refers to the extent to which a teacher believes that using a particular

AI tool will enhance their instructional effectiveness, while PEOU reflects how effortless the teacher expects the tool to be (Davis, 1998).

Recent work by Guo et al. (2024) validates an instrument tailored to measuring teachers' acceptance of AI in education. Their study confirms that PU and PEOU remain central to AI-related decision-making but also highlights the importance of contextual factors such digital confidence, institutional support, and ethical awareness. These insights are particularly relevant for pre-service teachers who are forming initial institutional exposure or structured training.

Recent developments in AI acceptance research have further expanded the explanatory power of TAM by highlighting the role of teacher self-efficacy in shaping technology acceptance. While the original TAM conceptualizes perceived usefulness and perceived ease of use as the primary determinants of behavioral intention (Davis, 1989), more recent AI-specific adaptations argue that teachers' confidence in their ability to use AI technologies significantly influences these perceptions. Guo, Shi and Zhai (2024), through the development and validation of the Technology Acceptance of Artificial Intelligence (TAAI) model, demonstrate that AI self-efficacy strengthens teachers' perceptions of AI as both useful and easy to use, thereby increasing their willingness to integrate AI into educational practice. Their findings suggest that teachers who believe they possess the knowledge and capability to use AI are more likely to perceive AI positively and express stronger intentions to adopt it. Although the present study does not adopt the TAAI model as its primary theoretical framework, these findings reinforce the importance of self-efficacy as a psychological construct underpinning AI acceptance and provide additional support for the continued application of TAM in AI-related educational research.

Within the context of this study, teacher self-efficacy is understood as pre-service teachers' confidence in their ability to learn, evaluate and use AI tools to support Life Sciences teaching. This extends beyond confidence in operating digital technologies to include confidence in selecting appropriate AI tools, interpreting AI-generated outputs, and making pedagogically informed decisions regarding their classroom use. Self-efficacy therefore contributes directly to perceived ease of use, as teachers who feel capable of engaging with AI are more likely to perceive AI technologies as manageable rather than complex. It also contributes indirectly to perceived usefulness because teachers who are confident in their technological abilities are

more likely to recognize the potential of AI to enhance teaching and learning. Consequently, self-efficacy represents an important psychological mechanism through which pre-service teachers develop preparedness for AI integration.

Moreover, Ma and Liu's (2005) meta-analysis supports the reliability of TAM across diverse educational and technological contexts, reinforcing its suitability for studies exploring educational innovations such as AI. For this research, TAM is used not only to examine pre-service teachers perceived benefits and suitability of AI tools but also to identify potential barriers to adoption, such a lack of training or perceived relevance to Life Sciences content. The model helps to interpret pre-service teachers' perceptions of AI tools, including whether they see such tools as enhancing Life Sciences teaching and whether they believe these tools are accessible and manageable. TAM therefore provides a conceptual basis for understanding the affective and cognitive aspects of perceived preparedness and anticipated adoption (Davis, 1989).

By focusing on attitudinal and psychological preparedness, TAM provides a critical interpretive lens for understanding how pre-service teachers' perceptions of usefulness, perceived ease of use and self-efficacy collectively influence their behavioral intention to integrate AI. Within this study, TAM therefore explains the motivational dimension of AI preparedness by accounting for the beliefs and perceptions that shape participants' willingness to adopt AI in their future Life Sciences classrooms. However, willingness alone does not necessarily indicate preparedness for effective classroom implementation. For this reason, TAM is complemented by the TPACK framework, which explains the pedagogical and technological knowledge required to translate behavioral intention into meaningful AI integration within subject teaching.

2.8.2 Technological Pedagogical Content Knowledge

Then second framework that informs this study Is the TPACK model, developed by Mishra and Koehler (2006). TPACK extends Shulman's (1986, 1987) notion of Pedagogical Content Knowledge (PCK) by integrating Technological Knowledge (TK) domain. It describes the intersection of three knowledge domains necessary for effective technology integration in

teaching: Content Knowledge (CK), understanding subject matter, in this case, Life Sciences, Pedagogical Knowledge (PK), knowledge of effective teaching methods and strategies, and Technological Knowledge (TK), understanding of digital tools and how to use them in instruction (Mishra & Koehler, 2006).

Although TPACK comprises seven interrelated knowledge domains, the present study does not examine each component equally. Because the participants are third and fourth-year pre-service Life Sciences teachers, the study does not seek to evaluate their disciplinary content knowledge, which is expected to have been developed through their teacher education programme. Instead, the study primarily draws on Technological Knowledge (TK), Pedagogical Knowledge (PK), and particularly Technological Pedagogical Knowledge (TPK), as these domains most directly relate to participants' perceived preparedness to integrate AI into Life Sciences teaching. Within this context, TPACK provides a conceptual framework for understanding how pre-service teachers anticipate selecting, using, and pedagogically applying AI tools to support curriculum delivery, learner engagement, and conceptual understanding.

Including TPACK is therefore appropriate given the study's focus on preparedness, as it provides a theoretical lens for understanding pre-service teachers' perceived capability to integrate AI within Life Sciences teaching. It acknowledges that AI integration is not just a matter of acceptance (as TAM focuses on) but also involves teacher's sense making about how AI aligns with disciplinary goals, classroom strategies, and digital competencies. TPACK complements TAM by offering a pedagogical lens on how participants envision the practicalities of using AI to teach Life Sciences, even if they are still developing in all three knowledge areas.

2.8.3 TAM-TPACK Integration

The TAM and the adapted AIPK framework are not employed as independent theories but as complementary theoretical lenses that explain different dimensions of pre-service teachers' preparedness for AI integration in Life Sciences education. While TAM explains how teachers' beliefs about the usefulness and ease of using AI shape their behavioral intentions towards AI adoption (Davis, 1989), behavioral intention alone does not necessarily indicate that teachers are prepared to implement AI effectively in practice. Recent scholarship has highlighted the complementary relationship between technology acceptance and teachers' pedagogical

knowledge, suggesting that these perspectives provide a more comprehensive understanding of technology integration than either framework alone (Deb & Sofal, 2026).

The adapted AIPK framework extends this understanding by providing the pedagogical knowledge dimension of preparedness. It explains how preservice teachers anticipate integrating AI into Life Sciences teaching. Rather than viewing AI adoption solely as a matter of acceptance, the framework emphasizes teachers' perceived capability to apply AI pedagogically in ways that support curriculum delivery, learner engagement, and conceptual understanding. As discussed in Section 2.8.2, this study places particular emphasis on AIK, PK, and AIPK because these domains most directly reflect pre-service teachers' perceived preparedness to integrate AI into teaching, while the remaining domains are retained within the adapted framework but are not examined as independent analytical constructs.

Accordingly, preparedness for AI integration is conceptualized in this study as more than teachers' acceptance of AI technologies. Rather, preparedness emerges through the interaction between pre-service teachers' perceptions of AI, their behavioral intentions towards AI adoption, and their perceived pedagogical capability to integrate AI meaningfully within Life Sciences education. The integrated TAM–AIPK framework therefore provides a comprehensive explanation of AI preparedness by recognizing that positive perceptions of AI are necessary but insufficient without the pedagogical knowledge required for effective classroom implementation. By integrating these complementary perspectives, the framework provides a holistic understanding of the motivational and pedagogical factors that shape pre-service teachers' preparedness to integrate AI into their future professional practice.

The integrated framework informed the development of the research questions, the design of the qualitative questionnaire and semi-structured interview protocol, and the thematic analysis used to interpret participants' perceptions, anticipated pedagogical practices, and preparedness for AI integration. By combining the motivational perspective of TAM with the pedagogical perspective of the adapted AIPK framework, the study provides a holistic understanding of how pre-service Life Sciences teachers perceive and prepare for the integration of AI within their future professional practice.

2.9 Chapter Summary

This chapter has provided a comprehensive review of the literature and theoretical foundations surrounding the integration of AI in education, with a specific focus on Life Sciences within the South African context. The review highlighted that while AI offers transformative potential for subject-specific pedagogy particularly as a tool for "visualizing the invisible" abstract biological processes like meiosis and DNA replication, its adoption in Africa remains slow due to systemic infrastructural limitations.

The literature suggests that pre-service teachers often view AI with a mix of enthusiasm and ethical pragmatism; they recognize its power to personalize learning and reduce instructional load, yet they harbour significant concerns regarding learner dependency, academic integrity, and the digital divide. Furthermore, the review identified a critical knowledge-practice gap, noting that personal digital fluency with AI tools does not inherently translate into professional pedagogical preparedness.

By grounding this study in the TAM and the TPACK framework, this chapter established that teacher preparedness is a multidimensional construct shaped by PU and the complex interplay between content, pedagogy, and technology. Despite the high perceived value of AI, Perceived PEOU is significantly moderated by contextual realities such as load shedding and resource disparities in South African schools. Collectively, these insights reveal a significant research gap concerning subject-specific AI integration for future science educators, providing a clear rationale for the qualitative exploratory case study and the research methodology detailed in the following chapter.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the methodological approach adopted to investigate Life Sciences pre-service teachers' perceptions and preparedness for the integration of Artificial Intelligence. It details the research design, research paradigm, sampling strategy, data collection methods, and data analysis procedures employed in the study. The chapter further discusses ethical considerations, issues of trustworthiness, and the limitations inherent in the chosen design. By providing a transparent account of how the study was conducted, this chapter establishes the credibility and rigour of the research process.

3.2 Research Design

This study was situated within the interpretivist paradigm**,** which assumes that reality is socially constructed and that individuals develop meanings through their interactions with their social and educational environments (Cohen et al., 2018). Rather than seeking a single objective reality, interpretivism recognizes that participants may hold multiple realities shaped by their personal experiences, beliefs, and contextual circumstances. This paradigm was particularly appropriate for the present study because it sought to understand how pre-service Life Sciences teachers construct meaning around artificial intelligence (AI) as an emerging educational technology. Participants' perceptions of AI, their beliefs about its educational value, and their preparedness to integrate AI into future teaching are not viewed as fixed or universal but as socially constructed through teacher education, teaching practice, prior experiences with technology, and the broader South African educational context. An interpretivist perspective therefore enabled the study to explore these subjective meanings and understand how participants interpreted AI within their own professional and educational experiences.

This study adopted a qualitative research approach because the purpose was to explore and interpret participants' subjective perceptions, beliefs, and anticipated practices regarding AI integration rather than to measure predetermined variables or test causal relationships (Creswell, 2014). A qualitative approach was therefore appropriate because it allowed participants to describe their experiences, expectations, concerns, and understandings of AI in their own words, providing rich contextual insights that would not have been captured through

quantitative measures alone. Since AI integration in South African Life Sciences education remains an emerging phenomenon, qualitative inquiry was particularly suitable for exploring how pre-service teachers make sense of AI within their teacher education experiences and how these meanings influence their perceived preparedness for future classroom practice. The use of open-ended questionnaires, supplemented by semi-structured interviews, enabled the collection of detailed descriptions that reflected the complexity of participants' perspectives and experiences.

Within this approach, the study employed an exploratory case study design. A case study is defined as an in-depth investigation of a bounded system within its real-life context (Yin, 2018). In this study, the bounded case comprised third and fourth-year pre-service Life Sciences teachers enrolled in a single teacher education programme at one South African university. The case was therefore bounded by the institution, the teacher education programme, the participant group, and the phenomenon under investigation, namely pre-service teachers' perceptions and preparedness for AI integration in Life Sciences education. This design was appropriate because it enabled an in-depth exploration of AI preparedness within its authentic educational context rather than seeking statistical generalization beyond the case.

3.3 Participants and Sampling

The target population for this study was third- and fourth-year Bachelor of Education students majoring in Life Sciences at a South African university. These students were selected because they were at an advanced stage of their teacher education programme and had been exposed to curriculum planning, teaching practice, and the use of digital tools, making them relevant participants for investigating perceptions and preparedness regarding AI integration in Life Sciences education.

A combination of purposive and convenience sampling was employed. Purposive sampling was used because participants were intentionally selected based on characteristics relevant to the study, namely fourth and third-year pre-service Life Sciences teachers who had completed substantial teacher education coursework and teaching practice and were therefore able to provide informed perspectives on AI integration (Palinkas et al., 2015). Convenience sampling was incorporated because participants were recruited from a university that was readily

accessible to the researcher through course communication platforms and class announcements. Participation was voluntary, and only students who were willing to participate were included in the study.

A total of 38 participants completed the open-ended questionnaire. From this group, 4 participants were conveniently and purposively selected for follow-up semi-structured interviews based on their willingness to participate and the depth of their responses. This combination of participants supported both breadth and depth in the data, allowing for a more comprehensive understanding of the phenomenon under study.

While purposive sampling may introduce bias, this was mitigated by including participants with diverse experiences and perspectives. The study acknowledges that the findings are context-specific and may not be generalizable beyond the institutional setting.

3.3.1 Demographics

To provide a clear overview of the study participants, two tables are presented. The first table summarizes the demographic characteristics and prior AI exposure of the interview participants, which are reported individually to support the use of participant-linked quotations in the findings chapter. The second table presents a summary of questionnaire participants, grouping responses according to levels of prior exposure to AI tools in order to highlight patterns across the dataset. A more detailed table containing individual participant information for the full questionnaire sample is included in the **appendix A** to ensure transparency while maintaining clarity and conciseness in the main text.

Table 1: Interview Participant Demographics (n = 4)

Participant	Gender	Age range	Programme	Prior AI exposure
Pi1	F	21–25	B.Ed.	Moderate
Pi2	M	21–25	B.Ed.	Limited
Pi3	F	21–25	B.Ed.	Moderate
Pi4	F	26–30	B.Ed.	Limited

Prior AI exposure referred to participants' self-reported experience using AI technologies before participating in the study. As participants had not received formal instruction on AI within their teacher education programme, their exposure was primarily acquired through

independent personal use and academic activities, including supporting coursework, understanding concepts, completing assignments, lesson planning, and information searching. Participants' prior AI exposure was categorized as *moderate* when their responses indicated engagement with AI across multiple contexts (e.g., studies, teaching practice, and personal life) or for a variety of educational purposes using one or more AI tools. Exposure was categorized as *limited* when participants reported occasional or narrowly focused use of AI, typically confined to a single context or purpose, such as obtaining explanations, supporting assignments, or searching for information.

Table 2: Summary of Questionnaire Participants' Prior AI Exposure (n = 38)

Level of AI exposure	Description	Number of participants
No exposure	No meaningful prior use reported	2
Limited	Basic use (e.g., definitions, simple searches, minimal tools)	7
Moderate	Use across studies/teaching (assignments, lesson ideas, simplification)	21
Broad	Multiple tools + multiple contexts (teaching, personal, academic, varied tools)	8
Total		**38**

All participants were fourth and third-year Bachelor of Education (Life Sciences) students who had completed the teaching practice requirements prescribed by their teacher education programme. Consequently, teaching practice was treated as a common characteristic of the participant group rather than an individual participant variable and is therefore not reported separately in Table 1 and Table 2.

3.4 Data Collection

Data for this study were collected in two qualitative phases using an open-ended questionnaire followed by semi-structured interviews. These methods were selected because they enable the exploration of participants' subjective understandings, experiences, and expectations in their own words, thereby generating rich and contextually grounded insights (Cohen et al., 2018). Employing two complementary methods also supported methodological triangulation, enhancing the depth and credibility of the findings.

The first phase of data collection involved a self-administered electronic questionnaire distributed via Google Forms to third and fourth year pre-service Life Sciences teachers in one South African university. The questionnaire consisted of eighteen questions, including three background questions and fifteen open-ended questions organized into five sections: (1) perceptions of AI in education, (2) anticipated use of AI in Life Sciences teaching, (3) preparedness and knowledge, (4) contextual and institutional factors, and (5) final reflections. This structure enabled the researcher to progress from participants' prior experiences with AI to their perceptions, anticipated classroom practices, perceived preparedness, contextual influences, and recommendations for future AI integration in teacher education.

Table 3: Overview of data collection instruments

Instrument	Purpose in the study	Key focus areas	Example item/prompt
Open-ended questionnaire	To generate initial, broad perceptions of AI in Life Sciences	Perceptions of AI, anticipated uses, preparedness, constraints	"Describe how you currently understand the role of AI in Life Sciences teaching."
Semi-structured interviews	To explore and deepen initial questionnaire responses	Teacher identity, ethical concerns, contextual realities	"Can you give an example of how you might use an AI tool to teach a difficult Life Sciences concept?"

3.4.1 Open-ended Questionnaire

The first phase involved a self-administered open-ended questionnaire distributed electronically to thirty-eight fourth and third-year pre-service Life Sciences teachers via email using Google Forms. The questionnaire was designed to elicit detailed reflections on participants' understandings of AI in education, their anticipated pedagogical use of AI in Life Sciences classrooms and their perceived preparedness and confidence to integrate AI into teaching practice.

Open-ended questions allowed participants to provide in-depth and unrestricted responses, thereby facilitating the emergence of unanticipated insights and personally meaningful interpretations (Oppenheim, 2000).

The questionnaire was theoretically informed by the TAM and the adapted AIPK framework. Questions informed by TAM explored participants' perceptions of the educational value of AI, its perceived ease of use, and their behavioral intentions towards integrating AI into future Life Sciences teaching. Building on these motivational dimensions, questions informed by the adapted AIPK framework explored participants' AIK, PK, and AIPK by examining how they anticipated using AI to support the teaching of Life Sciences concepts, learner engagement, and pedagogically meaningful classroom practice. Together, these complementary frameworks enabled the questionnaire to explore both the motivational and pedagogical dimensions of pre-service teachers' preparedness for AI integration.

Prior to full distribution, the questionnaire was piloted with three pre-service teachers who were not part of the main study. The pilot study revealed that while the initial questionnaire was comprehensive, its length led to respondent fatigue, which could have compromised the depth of the open-ended responses. To address this, the researcher consolidated overlapping questions, specifically merging general perceptions of AI with subject-specific anticipated use. This refinement ensured the instrument could be completed within the 20-minute timeframe indicated in the participant information sheet.

3.4.2 Semi-structured Interviews

The second phase involved individual semi-structured interviews with four participants who had completed the questionnaire and indicated their willingness to participate in a follow-up interview. Individual interviews were selected to provide participants with an opportunity to discuss their experiences and views openly while enabling the researcher to probe questionnaire responses, clarify meanings, and explore emerging ideas in greater depth. Each interview lasted approximately 20–30 minutes and followed a flexible interview guide informed by participants' questionnaire responses while maintaining consistency across all interviews.

The interview guide was informed by the integrated TAM and the adapted AIPK framework. Rather than examining these frameworks independently, the interviews were designed to explore how the motivational and pedagogical dimensions of preparedness for AI integration interacted within participants' anticipated teaching practice. Questions informed by TAM

explored participants' perceptions of the educational value of AI, their confidence in using AI, and the factors influencing their intention to integrate AI into future Life Sciences teaching. Building on these discussions, questions informed by the adapted AIPK framework explored how participants anticipated applying AI pedagogically, including how they envisioned using AI to teach Life Sciences concepts, support learner engagement, promote conceptual understanding, and integrate AI meaningfully into classroom practice.

Follow-up probing questions enabled participants to elaborate on their experiences, clarify their responses, and explain how contextual factors, including teacher education experiences, institutional support, school realities, and resource availability, shaped their perceived preparedness to integrate AI into their future teaching. This approach allowed the interviews to extend beyond participants' acceptance of AI by exploring the pedagogical reasoning underpinning their anticipated classroom use of AI.

Interviews were conducted exclusively via Microsoft Teams. Participants were informed that the use of video was optional and were permitted to keep their cameras off to ensure privacy and comfort. No personally identifying information, including participants' names, was collected during the interviews in order to maintain anonymity from the point of data generation. All interviews were audio-recorded after informed consent had been obtained, and recordings were transcribed and anonymized to protect confidentiality.

Although an interview guide was used to ensure consistency across participants, additional probing questions were asked where appropriate to clarify responses and obtain richer descriptions of participants' experiences, perceptions, and anticipated practices regarding AI integration.

3.5 Data Security

To ensure ethical handling and protection of participant information, all digital data, including questionnaire responses, recordings, and interview transcripts were stored in a password-protected electronic folder accessible only to the researcher and academic supervisor. No identifying information was stored alongside the research data.

Where digital communication platforms such as WhatsApp were used, participants were informed of potential privacy limitations associated with third-party applications. Communication via these platforms was restricted to interview scheduling and data collection only. Immediately after each interview, audio recordings were transferred to secure offline storage and deleted from the recording device to minimise risk of unauthorized access.

In accordance with institutional ethical guidelines, all research data will be retained securely for a period of five years following completion of the study, after which they will be permanently destroyed.

3.6 Data Analysis

This study employed reflexive thematic analysis, following Braun and Clarke's (2006) six-phase approach, to analyze the open-ended questionnaire responses and semi-structured interview transcripts. Thematic analysis was selected because it provides a flexible yet systematic method for identifying patterns of meaning across qualitative datasets while remaining consistent with the interpretivist paradigm underpinning this study.

The analysis was conducted manually using Microsoft Word and Microsoft Excel rather than qualitative data analysis software such as NVivo. Manual coding enabled close engagement with participants' responses throughout the analytical process, while digital tools were used to organize codes, categories, and themes systematically. The questionnaire responses were analyzed first to identify recurring patterns across the larger participant group. Following this initial analysis, the interview transcripts were analyzed using the preliminary coding framework while remaining open to new insights and perspectives that emerged through the richer interview data. The interview findings therefore served both to confirm patterns identified in the questionnaire data and to elaborate, clarify, and refine the developing themes.

The analysis adopted a hybrid inductive-deductive approach. During the initial stages, coding was primarily inductive, allowing codes to emerge directly from participants' responses without imposing predetermined categories. This ensured that participants' own meanings and experiences remained central to the analysis and allowed unexpected insights to emerge. Once the initial coding process had been completed, deductive analysis was undertaken by interpreting the developing themes through the theoretical lenses of the Technology

Acceptance Model (TAM) and the Technological Pedagogical Content Knowledge (TPACK) framework. TAM informed the interpretation of themes relating to perceived usefulness, perceived ease of use, and preparedness for AI adoption, whereas TPACK informed the interpretation of participants' technological, pedagogical, and technological pedagogical knowledge relating to AI integration within Life Sciences teaching. This hybrid analytical approach ensured that the findings remained grounded in participants' lived experiences while also being interpreted through established theoretical perspectives.

Table 4: Development of themes from initial codes

Representative data extracts	**Initial codes**	**Category**	**Final theme**
"Most of what I've known about AI has come from my own exploration, social media, or peers." "I mostly use ChatGPT to understand difficult concepts and assignments." "I have attended university conferences where AI was discussed."	Self-directed learning; Independent exploration; Peer learning; Informal AI exposure; Academic use of AI	Sources and forms of AI knowledge	**Theme 1: Forms of AI Preparedness**
"Training has been very limited… we were not shown how to integrate AI into lessons." "We have general ICT knowledge but not AI-specific training." "Universities should provide practical AI workshops."	Lack of formal AI training; Limited pedagogical guidance; ICT rather than AI preparation; Need for practical training	Institutional preparation and pedagogical development	**Theme 2: The Theory–Practice Gap in AI Pedagogical Knowledge**
"I see AI as a support tool rather than replacing teachers." "AI can help me	AI as teaching support; Teacher-centered use;	Perceived pedagogical role of AI	**Theme 3: AI as a Pedagogical Support Tool**

prepare lessons and explain difficult concepts." "AI should assist teachers, not replace professional judgement."	Efficiency; Professional assistance		
"Simulations help learners visualize processes that cannot be seen." "Animations can simplify DNA replication and protein synthesis." "AI can explain abstract concepts at different learner levels."	Visualization; Simulations; Animations; Simplifying abstract concepts; Differentiated explanations	Supporting conceptual understanding	**Theme 4: AI for Visualizing Abstract Life Sciences Concepts**
"Learners may become too dependent on AI.""AI could reduce learners' critical thinking." "Students might copy AI answers without understanding."	Over-reliance; Reduced critical thinking; Dependency; Passive learning	Risks to learner engagement	**Theme 5: Learning Risks and Cognitive Concerns**
"AI may generate incorrect information." "Teachers must verify AI-generated content." "Ethical use of AI is very important."	AI inaccuracies; Hallucinations; Verification; Ethical responsibility; Professional judgement	Ethical and professional considerations	**Theme 6: Professional Identity and Ethical Concerns**
"Some schools do not have internet or electricity." "Many rural schools cannot support AI technologies." "Resource inequalities will affect AI implementation."	Infrastructure limitations; Digital divide; Connectivity; Resource inequality	Contextual barriers	**Theme 7: Contextual Constraints and Inequality**
"Universities should include AI in teacher education.	Curriculum reform; Practical workshops;	Capacity building and	**Theme 8: Lack of Institutional**

"Hands-on workshops would improve confidence." "Teachers need continuous professional development on AI."	Institutional support; Professional development	institutional support	**Support and Training**
"I would use AI to generate quizzes and assessments." "AI could help personalize learning." "I would use AI to explain difficult topics and design learning activities."	Assessment support; Personalized learning; Lesson planning; Differentiated instruction; Content explanation	Classroom applications of AI	**Theme 9: Anticipated Pedagogical Uses of AI**

While thematic analysis provided a flexible and systematic method for identifying patterns of meaning across participants' responses, it is not without limitations. Thematic analysis relies on the researcher's interpretive judgement during coding and theme development, which may introduce subjectivity into the analytical process (Braun & Clarke, 2006; Nowell et al., 2017). Furthermore, because thematic analysis prioritises depth of meaning within a specific context, the findings are not intended to be statistically generalisable beyond the study setting. To mitigate these limitations, this study employed reflexive journalling, an audit trail, and member checking to enhance transparency, credibility, and trustworthiness of the analysis.

In later stages, the themes were interpreted through the lens of TAM and the TPACK framework. TAM was used to interpret themes related to participants' perceived usefulness and perceived ease of use of AI in Life Sciences classrooms, while TPACK assisted in analysing how participants envisioned the integration of AI across technological, pedagogical, and subject-matter dimensions. This analytical strategy enabled the study to balance participants' lived experiences with theoretical grounding, thereby enhancing the authenticity and interpretive depth of the findings.

However, TAM has been critiqued for its limited attention to contextual, social, and institutional factors influencing technology adoption (Bagozzi, 2007; Venkatesh et al., 2003). In recognition of these limitations, TAM in this study was not applied as a predictive model

but rather as an interpretive lens, complemented by TPACK to account for pedagogical and content-specific dimensions of AI integration. This combined theoretical framing strengthened the explanatory depth of the analysis within the context of Life Sciences teacher education.

3.7 Rigour and Trustworthiness

To enhance rigour and trustworthiness, multiple strategies were employed in alignment with recognised qualitative research standards (Lincoln & Guba, 1985; Nowell et al., 2017). Credibility was strengthened through member checking**,** defined as the process of obtaining participant feedback to verify the accuracy and credibility of the researcher's interpretations (Birt et al., 2016). Because the questionnaire was completed anonymously, member checking was conducted only with the four participants who participated in the semi-structured interviews. Following transcription, each participant was invited to review their interview transcript alongside the corresponding interview recording to verify that their responses had been captured accurately and that their intended meanings had been preserved. Participants confirmed the accuracy of the transcripts and did not request any amendments or corrections. This process enhanced the credibility of the study by ensuring that participants' views were represented faithfully before the final stages of coding and thematic interpretation.

Method triangulation was employed by analyzing and comparing data obtained from the open-ended questionnaires and the semi-structured interviews (Creswell & Clark, 2011). Questionnaire responses provided broad insights into participants' perceptions, anticipated practices, and preparedness for AI integration, while the interviews enabled a deeper exploration and clarification of these perspectives. Patterns that were consistent across both datasets strengthened the credibility of the findings, whereas divergent responses were retained and examined to provide a richer and more nuanced understanding of participants' perceptions and preparedness.

An audit trail was maintained throughout the research process as a transparent and traceable record of analytical decisions, coding processes, theme development, and reflective memos (Lincoln & Guba, 1985). This process enhanced dependability by enabling others to trace the progression and rationale of the analysis.

Reflexivity was practised through the maintenance of a reflective research journal documenting the researcher's evolving assumptions, positionality, and potential biases (Berger, 2015). As a fellow student and occasional tutor within the same faculty as the participants, the researcher acknowledged that familiarity with the programme could both facilitate openness and introduce response bias linked to perceived academic expectations. Ongoing critical reflection was therefore undertaken to ensure that interpretations remained grounded in participants' accounts rather than researcher preconceptions.

3.8 Ethical Considerations

Ethical conduct in the research process was guided by the principles of respect for persons, beneficence, and justice as outlined in the Belmont Report (1979). Ethical clearance was obtained from the university of the Witwatersrand Human Research Ethics Committee prior to data collection.

Participation was voluntary, and all participants provided informed consent after receiving detailed information about the study's purpose, procedures, potential risks and benefits, and their right to withdraw at any time without penalty. Participants were informed that their responses would be anonymised and used solely for academic purposes. To ensure confidentiality and data security, all identifying information was removed, and pseudonyms were used in transcripts and reporting.

Digital data, including audio recordings and transcripts, were stored in a password-protected electronic folder accessible only to the researcher and academic supervisor. In accordance with institutional guidelines, research data will be retained securely for five years following completion of the study, after which they will be permanently destroyed. Interviews conducted via digital platforms proceeded only with explicit participant consent.

Participants were advised to use private and secure internet connections to minimise privacy risks. Audio recordings were transferred to secure offline storage immediately after each interview and deleted from recording devices. The use of WhatsApp was restricted to interview scheduling and logistical communication only. The study was conducted with integrity, cultural sensitivity, and full respect for participants' autonomy and digital safety.

3.9 Chapter summary

This chapter outlined the methodological approach adopted to investigate pre-service teachers' perceptions and preparedness for the integration of Artificial Intelligence in Life Sciences education. Grounded within an interpretivist paradigm, the study employed a qualitative exploratory case study design to enable an in-depth understanding of participants' experiences, perspectives, and meaning-making processes. This approach was appropriate for capturing the complexity of how pre-service teachers engage with emerging technologies within a specific Initial Teacher Education context.

The chapter detailed the processes of participant selection, data collection, and data analysis. Data were generated through open-ended questionnaires and semi-structured interviews, allowing participants to articulate their views in their own words while providing opportunities for deeper exploration of emerging themes. Thematic analysis was used to identify patterns across the data, supported by systematic coding and careful interpretation. Ethical considerations, data management procedures, and strategies to ensure trustworthiness, including credibility, dependability, and transparency, were also addressed to strengthen the rigor of the study.

Overall, the chapter provided a transparent and coherent account of how the research was conducted, demonstrating alignment between the research design, methods, and research questions. By establishing the credibility and methodological integrity of the study, this chapter lays the foundation for the presentation and analysis of findings in the following chapter.

CHAPTER FOUR

DATA ANALYSIS AND FINDINGS

4.1 Introduction

This chapter presents the findings of the study, which explored pre-service Life Sciences teachers' perceptions and preparedness for the integration of AI in teaching within a South African university context. The findings are drawn from two data sources: open-ended questionnaire responses from thirty-seven participants and four semi-structured interviews. The use of both data sources enabled triangulation, allowing for a more comprehensive understanding of participants' views and experiences.

The data were analyzed using thematic analysis following Braun and Clarke's (2006) six-phase approach, as outlined in Chapter 3. Through this process, patterns of meaning were identified across the dataset and organized into themes aligned with the research questions. The analysis moves beyond simple description, highlighting not only what participants reported, but also how they made sense of AI in relation to their future teaching practice.

The chapter is structured according to the three research questions. Each section presents the relevant themes and sub-themes, supported by evidence from both the questionnaire responses and interview data. Participant voices are incorporated through direct quotations to illustrate key findings.

4.2 RQ1: Preparedness

> *What levels and forms of preparedness do third and fourth-year Life Sciences pre-service teachers report regarding AI integration in subject teaching?*

The findings reveal that pre-service Life Sciences teachers demonstrate emerging but uneven preparedness for the integration of AI in teaching. While participants generally expressed familiarity with AI tools and recognized their potential in education, their preparedness was largely shaped by self-directed learning, limited formal training, and uncertainty regarding pedagogical application. Across both questionnaire responses and interview data, two key dimensions of preparedness emerged: the forms through which knowledge was acquired, and

a clear gap between theoretical awareness and practical classroom integration.

4.2.1 Theme 1: Forms of AI Preparedness

Participants generally reported moderate and uneven levels of preparedness, characterized by a distinction between general familiarity with AI tools and limited pedagogical knowledge for classroom integration. While many participants expressed confidence in using AI at a basic or personal level, fewer felt adequately prepared to integrate it meaningfully into Life Sciences teaching.

A dominant pattern across the data was the reliance on informal and self-directed learning. Participants frequently indicated that their knowledge of AI had not been formally taught but instead developed through independent exploration, peer interaction, and exposure to digital environments. As one participant explained:

"Most of what I've known about the AI has come from my own exploration, social media or informal discussion with my peers or friends." (Pi1). This reflects a broader trend where AI knowledge is acquired outside formal teacher education structures, suggesting that pre-service teachers are navigating AI learning independently rather than through structured academic guidance.

Although participants demonstrated a willingness to engage with AI, their confidence was often described as partial or conditional. For example, one participant stated: "My confidence level in using AI is moderate… I feel like I can learn and adapt, but I don't have all the skills needed to use it responsibly and effectively." (Pi1). Similarly, another participant noted: "I would say I am prepared, but not to a larger extent… I still need more training." (Pi3). These responses highlight a key tension: participants feel technologically capable, yet pedagogically underprepared. Their confidence is therefore not rooted in formal competence, but rather in adaptability and exposure.

Furthermore, the findings suggest that participants differentiate between using AI as a tool and understanding how to teach with it. While many were comfortable using AI for tasks such as generating explanations or accessing information, they expressed uncertainty about how to apply these tools within structured teaching and learning processes. This theme reveals that

preparedness is not absent, but rather fragmented and informally developed, with pre-service teachers relying heavily on self-initiated learning pathways.

4.2.1.2 Conceptual ambiguity surrounding AI

"...using those tools, like the digital technologies... simulations... virtual labs..." (Pi2)

Although many participants demonstrated familiarity with AI and expressed confidence in discussing its educational potential, analysis of both the questionnaire and interview data revealed considerable conceptual ambiguity regarding what constitutes Artificial Intelligence. When asked how they would integrate AI into their future Life Sciences classrooms, participants frequently referred to simulations, virtual laboratories, virtual reality, multimedia technologies, and other forms of educational technology as examples of AI integration. While some of these technologies may incorporate AI, they are not inherently AI technologies. This suggests that participants often understood AI as a broad umbrella term for digital technologies rather than as a distinct category of educational technology.

“One that I would use is when it comes to meiosis or the cell divisions where we have some resources that are available like simulations online where the learners…” (Pi3)

This finding suggests that participants possessed general technological awareness but limited conceptual understanding of AI as a distinct educational technology. Rather than demonstrating AI-specific knowledge, many responses reflected broader understandings of technology-enhanced teaching. Consequently, participants' preparedness for AI integration appeared to be grounded in general digital competence rather than explicit AI literacy. Additionally, it aligns with the distinction made in Chapter Two between Artificial Intelligence, AI tools, educational technology, and digital technologies. While AI forms part of the broader educational technology landscape, the findings indicate that participants frequently used these concepts interchangeably. The inability to distinguish AI from other educational technologies may reflect limited explicit preparation for AI integration within the Initial Teacher Education programme, rather than simply a lack of exposure to technology.

4.2.2 Theme 2: The Theory–Practice Gap in AI Pedagogical Knowledge

While participants demonstrated general awareness of AI, a significant gap emerged between understanding AI conceptually and applying it meaningfully in Life Sciences teaching contexts. This theory–practice divide was consistently evident across both questionnaire and interview data, where participants highlighted the absence of formal training and the lack of structured guidance on how to integrate AI into actual classroom practice. As a result, pre-service teachers' knowledge of AI remains largely theoretical, with limited pedagogical grounding.

A dominant pattern across questionnaire responses was the acknowledgement that teacher education programmes have not adequately addressed AI integration. Many participants explicitly indicated that they had not received any formal training, with responses such as "No" (P12) and "I haven't had formal AI training yet… we touched on educational technology in general" (P18). This distinction between general ICT exposure and AI-specific preparation suggests that while pre-service teachers may be digitally literate, they are not being equipped with the specialized pedagogical knowledge required for AI integration. This gap was strongly reinforced in the interview data, where participants described their training as minimal and insufficient. For example, Pi1 explained that "the training has been very limited… we were not really shown how AI works, how to use it responsibly, or how to integrate it meaningfully into a science lesson," while Pi4 similarly noted that "we didn't necessarily get taught how to implement artificial intelligence within our teaching… most of the things we had to learn on our own." These responses highlight a consistent pattern across both data sources: AI knowledge is largely acquired independently, rather than through structured academic preparation.

In addition to the lack of formal training, participants expressed uncertainty about how to translate AI knowledge into classroom practice, particularly within the context of Life Sciences and curriculum requirements. Questionnaire respondents were able to articulate the kinds of support they would need, with one participant suggesting that "a full year course on the use of AI in classroom" (P7) would be necessary, while another emphasised the importance of "practical workshops where I can practice using AI for lesson planning" (P21). These responses indicate that participants are not only aware of their limitations but are also able to identify specific forms of pedagogical support required to bridge this gap. However, in the absence of

such training, their understanding of AI remains largely abstract, with limited connection to real classroom application.

This lack of pedagogical grounding also contributed to a reliance on general or decontextualized ideas about AI, rather than clear, curriculum-aligned teaching strategies. While participants could describe what AI is and what it can potentially do, they often struggled to demonstrate how it could be meaningfully integrated into structured teaching practices, particularly in relation to curriculum frameworks such as CAPS. Interview data further illustrated this uncertainty, with Pi1 noting the need to ensure that "whatever AI gives aligns with the curriculum… because as a teacher, I have the responsibility to ensure learners get correct information." This reflects an emerging awareness of the teacher's role as a mediator and validator of AI-generated knowledge, which adds an additional layer of complexity to AI integration in the absence of formal preparation.

"Not so confident because I haven't had any training on it" (P9)

This statement encapsulates the central issue within this theme: a lack of formal preparation directly constrains both confidence and pedagogical preparedness. Importantly, the data suggest that participants' reluctance is not rooted in resistance to AI, but rather in uncertainty about how to use it effectively and responsibly within teaching contexts.

Furthermore, the theory–practice gap was reinforced by participants' experiences during teaching practice, where opportunities to engage with AI in real classrooms were limited. While some exposure to digital tools was reported, AI-specific applications were largely absent, further restricting the development of practical competence. This indicates that the gap between theoretical awareness and classroom application is not only a result of university training but is also shaped by school-level contexts and resource availability.

To further understand participants' practical exposure to artificial intelligence, they were asked to indicate the AI tools they had previously encountered across their studies, teaching practice, and personal contexts. Their responses revealed varying levels of familiarity with different AI applications, ranging from general-purpose generative tools to more specialized platforms used for writing, simulations, and lesson preparation. This overview provides insight into the extent and nature of participants' prior interaction with AI technologies, offering a contextual

foundation for understanding the gap between awareness and pedagogical application identified in this study. The distribution of tools reported by participants is presented in Table 5 below.

Table 5: AI tools previously encountered by participants (n = 38)

AI Tool	**Number of Participants (n)**	**Main Contexts of Use**
ChatGPT	28	Studies, assignments, lesson planning, teaching practice, personal use
Meta AI	6	Personal use, general queries
Copilot	5	Academic writing, assignments
DeepSeek	4	Research, academic support
Perplexity AI	3	Research, information retrieval
Gemini AI	3	Study support, concept clarification
Grammarly	4	Academic writing, grammar checking
Notebook LM	2	Study support, summarization
Gamma AI	1	Presentation creation
Canva (AI features)	1	Teaching materials, visual aids
Quizizz (AI-supported)	1	Assessment, quizzes
PhET Simulations	1	Teaching simulations
Teachy	1	Lesson planning
Voice assistants (e.g., Google Assistant, Bixby)	3	Personal organization, reminders
General / unspecified AI	6	General academic and personal use

This theme demonstrates that preparedness is not simply a matter of awareness, but is deeply influenced by the lack of structured, practical, and subject-specific pedagogical training. While pre-service teachers are increasingly aware of AI and its potential, this awareness is not matched by the competence required to integrate it effectively into Life Sciences teaching, resulting in uncertainty, reliance on self-learning, and limited classroom application.

4.3 RQ2: Perceptions and Beliefs

What are pre-service teachers' perceptions and beliefs about the role and potential of AI in Life Sciences education?

The findings reveal that pre-service Life Sciences teachers hold largely positive yet critically balanced perceptions of artificial intelligence in education. Across both questionnaire and interview data, participants consistently recognised the potential of AI to enhance teaching and learning, particularly in relation to efficiency, accessibility, and conceptual understanding. However, these positive perceptions were accompanied by clear concerns regarding its impact on learning processes, teacher roles, and ethical use. This indicates that participants' beliefs are not uncritical or technologically deterministic but rather reflect a cautious and reflective stance towards AI integration. Four key themes emerged: AI as a pedagogical support tool, AI for visualizing abstract Life Sciences concepts, learning risks and cognitive concerns, and professional identity and ethical concerns.

4.3.1 Theme 3: AI as a Pedagogical Support Tool

A dominant perception across both questionnaire and interview data is that AI is best understood as a pedagogical support tool rather than a replacement for the teacher. Participants consistently described AI in functional and practical terms, emphasizing its ability to assist with teaching tasks, enhance understanding, and improve efficiency within the classroom. This framing positions AI as an augmentative technology, one that supports and enhances teaching rather than fundamentally transforming or replacing it.

Questionnaire responses revealed that participants associate AI with improving instructional efficiency and simplifying complex teaching tasks. For example, one respondent noted that AI provides "quick ways of assessing understanding and a better way of explaining complex terms" (P5), while another highlighted its usefulness in "creating quick assignments such as quizzes… to assess learner understanding" (P3). These responses suggest that pre-service teachers perceive AI as a tool that can streamline assessment processes, generate teaching resources, and support content delivery, thereby reducing the administrative and cognitive load associated with teaching.

In addition, participants emphasised the role of AI in enhancing productivity and saving time, particularly in contexts where teachers are required to manage large classes and extensive workloads. One respondent explicitly stated that "workload can be minimized" through the use of AI (P6), indicating that AI is seen not only as a pedagogical tool, but also as a means of improving teaching efficiency. This perception aligns with a broader understanding of AI as a tool that can handle routine or repetitive tasks, allowing teachers to focus more on meaningful engagement with learners.

These perceptions were strongly reinforced in the interview data, where participants articulated similar views, but with greater depth and contextualization. For instance, Pi1 described AI as:

"a support tool… something that can help both teachers and learners understand complex biological concepts in more meaningful and accessible ways"

This statement highlights two important aspects of participants' perceptions. First, AI is explicitly positioned as a support mechanism, rather than an autonomous teaching agent. Second, its value lies in its ability to enhance understanding and accessibility, particularly in relation to complex subject content. Similarly, Pi2 described AI as involving the use of tools such as "simulations" and "virtual labs" to "enhance student learning," further reinforcing the idea that AI functions as a facilitator of learning experiences rather than a replacement for teaching.

Pi4 similarly viewed AI as a pedagogical resource that supports, rather than replaces, the teacher. She explained that AI can assist learners in understanding abstract Life Sciences concepts through simulations and animations that make complex scientific processes more accessible. However, she emphasised that teachers remain responsible for guiding learning and ensuring that AI is used meaningfully within the classroom. This reinforces participants' shared perception that AI complements, rather than substitutes, professional teaching practice. Pi3 also described AI as supporting learner-centred teaching by enabling teachers to personalise learning according to learners' different abilities and learning needs. She explained that AI allows teachers to adapt explanations and activities while maintaining the teacher's central role in facilitating learning. This suggests that participants viewed AI as extending teachers' pedagogical capabilities rather than replacing their professional expertise.

Importantly, participants consistently rejected the notion that AI could replace the teacher. Instead, they emphasised that AI lacks the human qualities necessary for effective teaching,

such as relational engagement, emotional support, and contextual responsiveness. This perspective was evident in both data sources, where AI was repeatedly described as an assistant that provides support, ideas, and resources, but does not replace professional judgement. For example, one questionnaire respondent explained that AI should "guide me, give me ideas or refine my already existing ideas" (P14), indicating that the teacher remains the primary agent in the teaching process.

Furthermore, participants' perceptions of AI as a support tool are closely linked to an emerging understanding of the teacher's role as a facilitator of learning, rather than a sole transmitter of knowledge. Interview data suggest that AI may enable teachers to shift their focus from delivering content to supporting learner understanding, guiding inquiry, and fostering critical thinking. In this sense, AI is not perceived as diminishing the teacher's role, but rather as reconfiguring it in ways that emphasize facilitation, guidance, and learner-centered pedagogy.

This theme demonstrates that pre-service teachers conceptualize AI as a practical and supportive resource that enhances teaching efficiency, improves access to knowledge, and supports learner understanding, while still maintaining the centrality of the teacher in the educational process. This perception reflects a balanced and pragmatic view of AI, where its value is recognised, but its limitations are also clearly understood.

4.3.2 **Theme 4: AI for Visualising Abstract Life Sciences Concepts**

A prominent and discipline-specific perception emerging from the data is the role of AI in enhancing the visualization of abstract and complex Life Sciences concepts. Participants consistently identified this as one of the most valuable and practical applications of AI, particularly in relation to topics that are difficult to observe, conceptualize, or explain through traditional teaching methods. This theme reflects a strong alignment between the affordances of AI technologies and the epistemological demands of Life Sciences as a subject.

Across questionnaire responses, participants frequently described AI as a tool that can transform abstract biological processes into visual, interactive, and accessible representations. For example, one respondent explained that AI could be used to "create flow charts with images and illustrations to explain complex processes" (P2), while another highlighted the use of "virtual labs for learners to actively see a process take place but in a quick and efficient way" (P4). Pi4 similarly highlighted the value of AI-generated simulations and animations for

teaching microscopic biological processes such as mitosis and meiosis. She explained that these processes are difficult for learners to observe directly, and that AI enables them to visualize concepts "that cannot be seen with the naked eye." In addition, she suggested that AI-generated quizzes could provide learners with immediate feedback, helping teachers identify misconceptions more effectively.

Likewise, Pi3 emphasised that AI-supported simulations promote active learner engagement rather than passive memorization. She explained that learners are able to explore biological processes step by step, revisit difficult concepts, and interact with content at their own pace, thereby supporting deeper conceptual understanding. These responses indicate that pre-service teachers perceive AI as enabling a shift from static and text-based representations of knowledge towards more dynamic and multimodal forms of learning.

This perception is particularly significant within the context of Life Sciences, where many key concepts, such as DNA replication, protein synthesis, cellular respiration, and genetic processes, are inherently microscopic, abstract, and temporally dynamic. Traditional teaching approaches often rely on diagrams or verbal explanations, which may limit learners' ability to fully grasp these processes. In contrast, AI is perceived as offering opportunities to visualize processes in motion, thereby enhancing conceptual understanding.

The interview data provided even stronger and more detailed support for this perception, illustrating how participants envision the use of AI in concrete teaching scenarios. For instance, Pi1 described how AI could be used when teaching complex topics such as DNA structure and protein synthesis:

"Instead of learners just memorizing steps from the textbook, they could actually see the process unfolding… they can pause it and replay it and ask questions about specific stages"

This account highlights several important pedagogical affordances of AI. First, it enables learners to observe processes dynamically, rather than relying on static representations. Second, it allows for learner control over the pace of learning, as students can pause, replay, and revisit content. Third, it supports interactive engagement, where learners can question and explore different stages of a process. Together, these affordances suggest that AI can facilitate more active and inquiry-based learning experiences.

Similarly, Pi2 emphasised the use of AI simulations to model complex biological concepts, explaining that tools such as simulations can be used "to model… protein synthesis… so that learners can see". This reinforces the idea that AI is perceived as a means of making abstract scientific phenomena visible and experientially accessible, thereby bridging the gap between theoretical knowledge and learner understanding.

In addition to supporting visualization, participants also highlighted the role of AI in differentiating explanations and adapting content to learners' varying levels of understanding. For example, Pi1 explained that AI can generate explanations at different levels of complexity, allowing some learners to engage with simplified, analogy-based explanations, while others engage with more advanced scientific language. This suggests that AI is not only seen as a visualization tool, but also as a means of supporting inclusive and differentiated learning within diverse classrooms.

Furthermore, participants linked the use of AI for visualization to increased learner engagement and motivation. By making content more interactive and relatable, AI is perceived as helping learners move beyond rote memorization towards deeper conceptual understanding. This is particularly important in Life Sciences, where understanding processes and relationships is central to meaningful learning.

This theme demonstrates that pre-service teachers perceive AI as a powerful tool for bridging the gap between abstract scientific knowledge and learner comprehension. Through simulations, animations, and interactive representations, AI enables learners to visualize and engage with complex Life Sciences concepts in ways that are not easily achievable through traditional teaching methods. This positions AI not only as a technological innovation, but as a pedagogically valuable resource that aligns closely with the conceptual demands of the subject.

4.3.3 Theme 5: Learning Risks and Cognitive Concerns

Despite the largely positive perceptions of AI outlined in the previous themes, participants expressed significant concerns regarding its potential impact on learners' cognitive development and engagement with learning. These concerns were consistently evident across both questionnaire responses and interview data, suggesting that pre-service teachers do not

view AI as unproblematically beneficial, but rather as a tool that must be used cautiously to avoid undermining meaningful learning processes.

A dominant concern across questionnaire responses was the risk of over-dependence on AI, where learners may rely on AI-generated responses instead of engaging actively with content. Participants frequently associated this over-reliance with a decline in effort and independent thinking. For example, one respondent noted that learners "might depend on it too much and not develop their own thinking skills" (P11), while another highlighted that AI use "can sometimes give out wrong information and make learners rely on it" (P8). These responses indicate a shared perception that, without appropriate guidance, AI may encourage passive learning behaviors, where learners consume information rather than critically engaging with it.

Closely linked to this concern is the notion of cognitive laziness, where the availability of AI-generated answers reduces the need for learners to think deeply, analyze information, or solve problems independently. Participants suggested that this could be particularly problematic in Life Sciences, a subject that requires not only content knowledge, but also the development of analytical and problem-solving skills. The concern here is not simply about technology use, but about its potential to reshape learning habits in ways that weaken higher-order thinking.

Interview data provided deeper insight into these concerns, particularly in relation to how AI might affect learners' engagement with assessment tasks. For instance, Pi1 explained that learners may:

"Use AI to complete assignment without understanding the content… which will undermine the learning"

This statement highlights a critical tension between task completion and genuine understanding, where learners may produce correct answers without developing the underlying conceptual knowledge. This raises important implications for teaching and assessment, as it becomes increasingly difficult to determine whether learners' work reflects their own understanding or AI-generated responses.

Pi4 expressed similar concerns regarding learner dependence on AI, arguing that excessive reliance on AI-generated responses could reduce learners' willingness to think critically and

solve problems independently. She emphasised that AI should support learning rather than replace learners' own reasoning processes.

Participants also expressed concerns about academic integrity, particularly in relation to assignments and assessments. The ease with which AI can generate responses was seen as creating opportunities for learners to bypass the learning process altogether. As Pi1 further noted, this creates challenges for teachers in terms of evaluating learner performance, raising questions about "how to assess learners fairly" in contexts where AI use is widespread. This concern reflects a broader anxiety about the validity of assessment practices in AI-integrated classrooms.

Pi3 similarly stressed that learners should be encouraged to question and verify AI-generated information rather than accepting it uncritically. According to her, teachers have an important responsibility to help learners recognise that AI is a support tool whose responses require critical evaluation.

In addition to issues of dependency and academic integrity, participants also raised concerns about the potential for AI to reinforce misconceptions, particularly if learners accept AI-generated information uncritically. As noted in questionnaire responses, AI can sometimes provide inaccurate or oversimplified explanations (P8), which may lead learners to develop incorrect understandings of complex scientific concepts. This is particularly problematic in Life Sciences, where misconceptions can be difficult to correct once established.

"Learners might depend on it too much and not develop their own thinking skills" (P11)

This statement captures the central concern within this theme: that the benefits of AI may come at the cost of reduced cognitive engagement, if its use is not carefully guided and regulated.

Importantly, participants did not reject AI outright based on these concerns. Instead, their responses suggest a recognition that these risks can be mitigated through responsible and structured use. Interview data indicate that teachers see themselves as playing a key role in guiding learners on how to use AI critically, rather than passively accepting its outputs. This suggests that the perceived risks are not inherent to AI itself but are contingent on how it is integrated into teaching and learning practices.

Overall, this theme highlights that pre-service teachers hold a cautious and critical perspective on AI, recognizing its potential to both support and undermine learning. While AI is seen as a valuable tool, participants emphasize the need for careful implementation to ensure that it promotes, rather than diminishes, critical thinking, conceptual understanding, and academic integrity.

4.3.4 Theme 6: Professional Identity and Ethical Concerns

In addition to concerns about learners, participants expressed important considerations related to their own roles, responsibilities, and professional identity as future Life Sciences teachers in AI-integrated classrooms. Across both questionnaire and interview data, participants demonstrated an awareness that the integration of AI is not only a pedagogical issue, but also a professional and ethical one, requiring careful judgement, accountability, and responsibility.

A key issue raised across the data was the accuracy and reliability of AI-generated information. Participants recognised that AI could produce responses that appear correct and authoritative, yet may contain inaccuracies, oversimplifications, or content that is not aligned with curriculum requirements. This concern was evident in questionnaire responses, where participants noted that AI "can give out wrong information (misconceptions)" (P8). This perception was further reinforced in the interview data, where Pi1 explained:

"AI can produce information that sounds correct, but may not always be accurate… I will need to double check what it is generating"

This reflects a clear understanding of the teacher's role as a knowledge validator, responsible for critically evaluating AI-generated content before it is used in teaching. Participants acknowledged that effective AI integration requires not only technological skills, but also strong content knowledge and critical judgement, in order to ensure that learners are not exposed to incorrect or misleading scientific information.

In addition to concerns about accuracy, participants raised important questions regarding professional dependency on AI. There was a shared concern that teachers themselves may become overly reliant on AI tools for lesson planning, assessment, and content generation, potentially undermining their own pedagogical development. As Pi1 cautioned:

“Some teachers might rely too much on AI instead of developing their own pedagogical skills”

This concern was echoed in the questionnaire data, where participants emphasised the importance of maintaining professional responsibility and not allowing AI to replace teacher thinking and decision-making. Interview data further deepened this concern, with Pi2 noting that some educators may simply “take assignments from AI… even the memo from AI,” which “might compromise the validity” of teaching and assessment practices. These responses highlight a perceived risk that AI could encourage professional complacency, where teachers rely on automated outputs rather than engaging in thoughtful pedagogical planning.

Pi4 reinforced these contextual concerns by explaining that many South African schools continue to experience unreliable electricity, limited internet connectivity, and inadequate access to learner devices. Nevertheless, she suggested that teachers could still benefit from AI by using it during lesson preparation to develop quizzes, explanations, and teaching resources, even where learners cannot access AI directly. This illustrates participants' attempts to reconcile the pedagogical potential of AI with the realities of resource-constrained classrooms.

Participants also reflected critically on the implications of AI for professional integrity and accountability, questioning the extent to which teaching remains authentic when heavily mediated by AI. This is captured in the concern raised by Pi3:

“Are teachers really doing their job, or just using AI for compliance?”

This statement reflects a deeper tension between efficiency and professionalism, where the use of AI must be balanced with the responsibility to engage meaningfully in the teaching process. It suggests that participants are not only concerned with how AI affects learners, but also with how it may reshape what it means to be a competent and responsible teacher.

Beyond issues of professional practice, participants also raised ethical concerns related to learner protection and data privacy, particularly when using online AI platforms. There was an awareness that encouraging learners to use AI tools may expose them to risks related to data security and digital safety. As highlighted in the interview data, teachers carry a “professional responsibility to protect learners,” especially when integrating technologies that operate beyond the controlled classroom environment. This indicates that participants recognize the

need for ethical awareness and responsible implementation, beyond purely instructional considerations.

Finally, participants emphasised the importance of preserving the human dimension of teaching, including relationships, emotional support, and contextual responsiveness. AI was not perceived as capable of replicating these aspects, reinforcing the view that teaching is inherently relational and cannot be reduced to automated processes. Instead, AI was seen as a tool that should support teaching, while the teacher remains central as a guide, mentor, and facilitator of learning.

Overall, this theme reflects a nuanced and professionally grounded understanding of AI integration, where pre-service teachers recognize both the opportunities and the responsibilities associated with its use. Their perceptions highlight that successful AI integration is not simply about adopting new technologies, but about maintaining professional judgement, ethical awareness, and pedagogical integrity, positioning teachers as active agents in regulating and mediating the role of AI in education.

4.4 RQ3: Factors Influencing Preparedness

> *How do experiences within the ITE programme and broader institutional context shape pre-service teachers' preparedness to integrate AI?*

The findings indicate that pre-service teachers' preparedness to integrate AI is not determined solely by their individual knowledge or perceptions, but is significantly shaped by institutional structures and contextual realities. Across both questionnaire responses and interview data, participants consistently highlighted that their ability to meaningfully integrate AI into Life Sciences teaching is influenced by factors such as school infrastructure, access to digital resources, and the nature of their teacher education programme.

While many participants expressed willingness and interest in using AI, their preparedness was often constrained by external conditions beyond their control, particularly within the South African context. These findings suggest that preparedness for AI integration is not simply a matter of individual competence, but is deeply embedded within broader systemic, institutional, and socio-economic contexts. Three key themes emerged: (1) contextual

constraints and inequality, (2) lack of institutional support and training, and (3) anticipated pedagogical uses of AI.

4.4.1 Theme 7: Contextual Constraints and Inequality

A dominant factor influencing preparedness is the presence of contextual constraints, particularly in relation to infrastructure, access to technology, and broader socio-economic inequalities. Participants consistently highlighted that the feasibility of AI integration is heavily dependent on material conditions within schools, including access to devices, stable internet connectivity, and reliable electricity.

Questionnaire responses revealed that many participants are aware of the practical limitations of implementing AI in under-resourced environments. For example, one respondent noted that AI use is limited by "lack of resources and access to devices" (P10), while another emphasised challenges related to "internet access and data availability" (P16) . These responses indicate that even when teachers are willing to use AI, structural barriers may prevent its effective implementation.

These concerns were strongly reinforced in the interview data, where participants provided more detailed accounts of how contextual realities shape their preparedness. For instance, Pi1 described how differences between schools impacted their ability to engage with AI during teaching practice: "some schools had smart boards and internet access, while others had none… especially in rural or township schools" This highlights the uneven distribution of resources across schools, which directly affects opportunities to experiment with and integrate AI tools in real classroom settings. Similarly, Pi2 noted that the use of AI is dependent on connectivity, explaining that without "devices and internet connectivity… that might hinder the AI integration" . This reinforces the idea that AI integration is contingent on technological infrastructure, which is not uniformly available across educational contexts.

Participants also demonstrated an awareness of the broader digital divide, recognising that AI may inadvertently widen existing inequalities between well-resourced and under-resourced schools. This suggests that preparedness is not only an individual issue, but also a matter of equity and access, where some teachers and learners are better positioned to benefit from AI than others.

Overall, this theme highlights that contextual realities act as gatekeepers of AI integration, shaping what is possible in practice. Even when pre-service teachers possess positive perceptions and a willingness to use AI, their preparedness is significantly limited by infrastructural challenges and systemic inequalities.

4.4.2 Theme 8: Lack of Institutional Support and Training

Another key factor influencing preparedness is the limited support provided by teacher education programmes in relation to AI integration. Across both data sources, participants consistently reported that their university training had not adequately prepared them to use AI in pedagogically meaningful ways, particularly within the context of Life Sciences teaching.

Questionnaire responses revealed a clear demand for structured and practical training, with participants emphasising the need for more than general exposure to educational technology. For example, one respondent suggested that universities should offer “a full year course on the use of AI in classroom” (P7), while another highlighted the importance of “practical workshops where I can practice using AI for lesson planning” (P21) . These responses indicate that participants are not only aware of the gap in their training, but are also able to articulate specific forms of support needed.

This perception was strongly reinforced in the interview data, where participants described their training as largely theoretical and lacking practical application. Pi1 explained that university programmes tend to “prioritise theory… rather than the practical,” and emphasised the need for “hands-on training” and opportunities to actively engage with AI tools in teaching contexts . Similarly, Pi2 noted that universities “still need to do more” and suggested the introduction of structured training, such as short courses or modules focused on AI integration.

Pi4 similarly reported that AI had not been explicitly incorporated into her teacher education programme. She explained that most of her understanding had been developed independently because "we didn't necessarily get taught how to implement artificial intelligence within our teaching." As a result, she described herself as only "semi-prepared" to integrate AI into future classroom practice.

Pi3 echoed this experience, explaining that although her programme introduced digital technologies and open educational resources, learning how to use AI to simplify complex content or prepare lessons largely resulted from her own initiative rather than structured

university training. These accounts further reinforce participants' perception that current teacher education programmes have not adequately prepared them for AI integration.

Participants also highlighted the importance of subject-specific training, rather than general ICT instruction. There was a clear expectation that AI integration should be embedded within Life Sciences pedagogy, including applications such as virtual labs, simulations, and curriculum-aligned teaching strategies. This reflects a recognition that effective AI integration requires not only technological familiarity, but also pedagogical alignment with subject content and curriculum demands.

In addition, participants emphasised the need for training that addresses ethical use, data privacy, and responsible implementation, indicating that preparedness involves both technical and ethical competencies. This suggests that current teacher education programmes may be insufficiently addressing the complex and multidimensional nature of AI integration.

Overall, this theme demonstrates that preparedness is significantly constrained by the lack of structured, practical, and subject-specific institutional support, highlighting the need for curriculum reform within teacher education programmes to better prepare pre-service teachers for AI-integrated teaching.

4.4.3 Theme 9: Anticipated Pedagogical Uses of AI

Despite the constraints identified in the previous themes, participants demonstrated a clear ability to envision how AI could be meaningfully integrated into their future teaching practice. This indicates that while their current preparedness may be limited, they possess emerging ideas about the pedagogical potential of AI, particularly within the context of Life Sciences.

Questionnaire responses revealed that participants intend to use AI primarily for supporting teaching of complex topics, enhancing learner understanding, and improving engagement. For example, one respondent noted that AI could be used to “create interactive simulations of complex processes like gene expression or cellular respiration” (P4), while another highlighted its potential for “helping learners understand difficult topics in a simpler way” (P15) . These responses align closely with earlier findings related to visualisation and conceptual understanding.

Participants also expressed interest in using AI for differentiated instruction and personalised learning, recognising that learners have diverse needs, abilities, and learning speeds. AI was

seen as a tool that can provide multiple explanations, adapt content, and support individual learning pathways, thereby making teaching more inclusive and responsive.

Interview data provided further depth, illustrating how participants envision applying AI in specific teaching scenarios. For instance, Pi1 described using AI to generate different levels of explanation for learners, while also incorporating simulations and interactive tools to support understanding. Similarly, Pi2 emphasised the use of AI-generated prompts and simulations to guide learners towards deeper conceptual engagement, rather than simply providing answers.

Pi4 also anticipated using AI to generate formative quizzes, provide immediate feedback, and develop simulations that support learners' understanding of abstract Life Sciences concepts. Similarly, Pi3 envisaged using AI to personalise learning by adapting explanations and activities to learners' differing levels of understanding. Together, these responses suggest that participants viewed AI not merely as a source of information but as a pedagogical tool capable of supporting differentiated instruction, learner engagement, and formative assessment.

In addition, participants highlighted the potential of AI to support assessment and feedback, particularly in contexts where large class sizes make it difficult to provide immediate and individualised feedback. AI was seen as a tool that can generate questions, quizzes, and feedback, thereby enhancing formative assessment practices.

Overall, this theme demonstrates that pre-service teachers possess emerging pedagogical visions for AI integration, even in the absence of formal training. While their current preparedness is constrained by contextual and institutional factors, their ability to articulate meaningful uses of AI suggests a preparedness to engage with these tools, provided that adequate support and resources are made available.

4.5 Chapter Summary

This chapter presented the findings of the study, organized according to the three research questions and supported by both questionnaire responses and interview data. Through thematic analysis, a number of key patterns emerged regarding pre-service Life Sciences teachers' preparedness, perceptions, and preparedness for the integration of artificial intelligence in teaching.

Findings related to RQ1 revealed that participants possess basic and functional awareness of AI, largely developed through informal and self-directed learning. However, this knowledge

remains fragmented and predominantly non-pedagogical, with participants expressing moderate confidence and uncertainty about applying AI meaningfully within classroom contexts. A clear theory–practice gap was identified, highlighting the disconnect between general awareness of AI and its pedagogical integration in Life Sciences teaching.

In relation to RQ2, participants demonstrated largely positive yet critically balanced perceptions of AI. AI was widely viewed as a supportive pedagogical tool with the potential to enhance teaching efficiency and improve understanding of abstract concepts through visualization and interactivity. At the same time, participants expressed important concerns regarding cognitive dependence, academic integrity, and the ethical and professional responsibilities associated with AI use, reflecting a cautious and reflective stance towards its integration.

Findings from RQ3 further indicated that preparedness for AI integration is significantly influenced by institutional and contextual factors, rather than individual knowledge alone. Participants highlighted challenges related to infrastructure, access to technology, and unequal school contexts, as well as the lack of structured, subject-specific training within teacher education programmes. Despite these constraints, participants demonstrated an emerging ability to envision meaningful pedagogical uses of AI, particularly in relation to simulations, differentiation, and assessment.

In conclusion, findings suggest that while pre-service teachers are increasingly aware of AI and its potential in Life Sciences education, their preparedness to integrate it effectively remains limited by gaps in pedagogical preparation and contextual constraints. The next chapter builds on these findings by interpreting them in relation to existing literature and theoretical frameworks, in order to provide a deeper understanding of their implications for teacher education and practice.

CHAPTER FIVE

DISCUSSION

5.1 Introduction

This chapter presents a critical discussion of the findings reported in Chapter 4, interpreting them in relation to the study's research questions, existing literature, and the theoretical frameworks underpinning the study. The purpose of this chapter is to move beyond description and provide a deeper analytical understanding of pre-service Life Sciences teachers' preparedness, perceptions, and preparedness for the integration of artificial intelligence (AI) within the context of ITE.

The discussion is guided by AIPK framework and the TAM, which together provide a complementary lens for understanding both the knowledge dimensions and the behavioral factors influencing AI integration. As outlined in Chapter 2, AIPK is essential for effective teaching with technology (Mishra & Koehler, 2006), while TAM explains how perceived usefulness and perceived ease of use shape individuals' acceptance and adoption of emerging technologies (Davis, 1989). These frameworks are particularly relevant in this study, as they allow for an examination of how pre-service teachers' developing knowledge structures intersect with their attitudes, beliefs, and contextual experiences within ITE.

The findings of this study are discussed in relation to the three research questions. First, the discussion examines pre-service teachers' preparedness for AI integration, with particular attention to the distinction between general technological familiarity and pedagogical competence. Second, the chapter explores participants' perceptions and beliefs regarding the role and potential of AI in Life Sciences education, including both enabling and constraining perspectives. Third, the discussion considers the institutional and contextual factors shaping preparedness, highlighting the role of teacher education programmes and broader systemic conditions in influencing preparedness.

5.2 Discussion of RQ1: Preparedness for AI Integration

The findings of this study indicate that pre-service Life Sciences teachers demonstrate a form of partial and uneven preparedness for AI integration, characterized by general familiarity with AI tools but limited pedagogical competence in applying these tools within teaching contexts.

While participants reported confidence in using AI for personal and academic purposes, this confidence did not extend to structured classroom integration. This distinction reflects a broader pattern identified in the literature, where exposure to digital technologies does not necessarily translate into pedagogical preparedness (Mnguni, 2023; Viberg et al., 2023).

From the perspective of the adapted AIPK framework, these findings suggest that participants demonstrated elements of Artificial Intelligence Knowledge (AIK), particularly in relation to recognizing, accessing, and using AI tools such as generative AI platforms for information retrieval, lesson preparation, and content generation. However, this knowledge appeared to operate largely independently of pedagogical application. As discussed in Chapter Two, effective AI integration requires more than familiarity with AI technologies; it requires the ability to connect AI capabilities with appropriate pedagogical strategies to support meaningful teaching and learning (Mishra & Koehler, 2006; Deb & Sofal, 2026). Although participants generally recognised the potential educational value of AI, many struggled to explain how AI could be purposefully integrated into Life Sciences pedagogy. This limitation was further reflected in participants' tendency to conflate AI with broader digital technologies, including simulations, virtual laboratories, virtual reality, and other ICT tools. Such conceptual ambiguity suggests that participants possessed general digital competence but more limited AI-specific pedagogical knowledge, indicating that their preparedness for AI integration was constrained by an incomplete understanding of AI as a distinct educational technology.

This disconnect is further reflected in what the findings identified as a theory–practice gap, where participants are able to describe AI in general or functional terms but struggle to translate this understanding into pedagogical action. Similar patterns have been reported in previous research, where pre-service teachers demonstrate conceptual awareness of emerging technologies but lack the pedagogical grounding required for implementation (Caner & Aydin, 2021; Mnguni, 2024). This gap highlights a critical limitation within teacher education, where technological exposure is not sufficiently integrated with subject-specific pedagogical training. Importantly, the findings also indicate that participants' knowledge of AI is largely informal and self-directed, acquired through personal exploration, peer interaction, and digital environments rather than structured academic engagement. This aligns with existing literature suggesting that, in the absence of formal training, pre-service teachers often develop technological familiarity independently, which may support basic operational competence but

does not necessarily foster pedagogical integration (Aslan & Zhu, 2016; Viberg et al., 2023). While such self-directed learning reflects adaptability and openness to innovation, it also contributes to fragmented knowledge structures that lack coherence within disciplinary teaching contexts.

An unexpected finding emerging from this study was the extent to which participants conflated AI with broader digital technologies such as simulations, virtual laboratories, virtual reality, and multimedia resources. This conceptual ambiguity suggests that participants' positive perceptions of AI were not always underpinned by a clear understanding of AI as a distinct technological domain. Rather than indicating AI-specific preparedness, many responses reflected broader digital pedagogical competence. This finding reinforces the importance of explicitly developing AI literacy within Initial Teacher Education programmes before expecting meaningful AI integration in classroom practice.

The role of ITE emerges as particularly significant in shaping this pattern of preparedness. As highlighted in Chapter 2, teacher education programmes play a critical role in developing not only technical skills but also pedagogical reasoning and professional identity (Ertmer & Ottenbreit-Leftwich, 2010; Mnguni, 2023). However, the findings of this study suggest that AI is not yet meaningfully embedded within the ITE curriculum, particularly within Life Sciences methodology. As a result, participants are left to navigate AI independently, without guided opportunities to explore its pedagogical affordances, limitations, and alignment with curriculum frameworks such as CAPS.

When interpreted through the TAM, an interesting dynamic emerges. Participants demonstrated relatively high levels of perceived usefulness, recognizing AI's potential to support learning, simplify complex content, and enhance teaching efficiency. According to Davis (1989), such perceptions are typically associated with positive attitudes toward technology adoption. However, perceived ease of pedagogical use appeared significantly lower, largely due to lack of training and limited opportunities for practical engagement. This imbalance between perceived usefulness and perceived ease of use has been identified in extended TAM studies as a key barrier to actual technology adoption, particularly when external variables such as training and institutional support are insufficient (Venkatesh et al., 2003).

Crucially, participants did not interpret their limited pedagogical preparedness as a personal shortcoming, but rather as a structural limitation within their teacher education experience. This distinction is important, as it shifts the focus from individual deficit to systemic responsibility. As Ertmer and Ottenbreit-Leftwich (2010) argue, teacher beliefs and knowledge are shaped within broader institutional contexts, and meaningful technology integration requires intentional support, modelling, and curriculum alignment. In this study, the absence of such support appears to constrain the development of integrated knowledge structures necessary for AI-enhanced teaching.

In the context of Life Sciences education, this gap is particularly significant. As noted in the literature, the discipline requires the teaching of complex, abstract, and often invisible processes, which can benefit from technologies that support visualization and simulation (Mnguni, 2023; Viberg et al., 2023). However, without the integration of technological, pedagogical, and content knowledge, the potential of AI to support conceptual understanding remains underutilized. The findings therefore suggest that preparedness is not simply about awareness or exposure, but about the development of integrated, context-sensitive knowledge structures that enable meaningful classroom application.

This paper demonstrates that pre-service teachers' preparedness for AI integration is characterized by emerging technological familiarity, strong perceived usefulness, but limited pedagogical integration, resulting in a persistent theory–practice gap. Addressing this gap requires a shift within ITE from general technological exposure toward structured, subject-specific, and pedagogically grounded AI integration, enabling pre-service teachers to move from awareness to meaningful implementation.

5.3 Discussion of RQ2: Perceptions and Beliefs about AI in Life Sciences Education

The findings of this study reveal that pre-service Life Sciences teachers hold generally positive yet critically balanced perceptions of AI, recognizing its pedagogical potential while simultaneously expressing important concerns regarding its implications for teaching and learning. These perceptions are significant, as they function as cognitive filters through which technological innovations are interpreted and adopted, ultimately influencing future classroom practice (Ertmer & Ottenbreit-Leftwich, 2010; Mnguni, 2023).

A dominant pattern emerging from the findings is the positioning of AI as a pedagogical support tool rather than a replacement for the teacher. Participants consistently viewed AI as enhancing efficiency, supporting lesson preparation, and assisting in the explanation of complex concepts. This aligns with existing research indicating that pre-service teachers often conceptualize AI as a means of augmenting, rather than substituting, human teaching (Akkaya, 2016; Mnguni, 2024). Such perceptions are important, as they reflect an orientation toward human-centered pedagogy, where technology is integrated to support, rather than displace, pedagogical agency.

From a TAM perspective, these findings indicate strong perceived usefulness, with participants recognizing AI's value in improving teaching efficiency, supporting learner understanding, and enhancing access to information. According to Davis (1989), perceived usefulness is a key determinant of positive attitudes toward technology adoption and is often associated with increased intention to use technology in practice. In this study, participants' recognition of AI's benefits, particularly in relation to simplifying complex Life Sciences content, suggests a strong foundation for future adoption.

One of the most significant perceived benefits relates to AI's ability to support the visualization of abstract and microscopic Life Sciences processes, such as cellular mechanisms, genetics, and ecological systems. This finding is consistent with literature highlighting the potential of AI-driven simulations and visual tools to enhance conceptual understanding in science education (Mnguni, 2023; Viberg et al., 2023). Given that Life Sciences often involves processes that are not directly observable, the ability of AI to make the "invisible visible" represents a meaningful pedagogical affordance. This reinforces the idea that participants' perceptions are not only technologically oriented, but also discipline-specific, reflecting an awareness of how AI aligns with the epistemological demands of Life Sciences.

However, these positive perceptions are accompanied by critical concerns, particularly in relation to learning processes and cognitive development. Participants expressed apprehension about the potential for over-reliance on AI, suggesting that excessive dependence may undermine learners' critical thinking, problem-solving abilities, and independent reasoning. Similar concerns have been reported in previous studies, where AI is perceived as potentially

reducing cognitive engagement if not used appropriately (Pokrivčáková, 2023; Karataş & Yüce, 2024). These findings highlight that participants do not engage with AI uncritically, but rather demonstrate an emerging awareness of its pedagogical risks.

Concerns were also raised regarding academic integrity, particularly the potential for learners to misuse AI for completing assignments or assessments without genuine understanding. This reflects broader debates in the literature on generative AI, where issues of authorship, authenticity, and assessment validity have become increasingly prominent (Ng et al., 2021). Such concerns further illustrate that pre-service teachers are already grappling with the ethical complexities associated with AI integration, even prior to entering the profession.

In addition to learner-related concerns, participants reflected on the implications of AI for their own professional identity as teachers. While AI was generally not viewed as a threat to the teaching profession, there was recognition that over-reliance on AI could undermine pedagogical expertise and professional judgement. This aligns with research suggesting that emerging technologies can challenge traditional conceptions of teacher roles, requiring educators to reposition themselves as facilitators, mediators, and critical evaluators of knowledge (Mnguni, 2024; Ertmer & Ottenbreit-Leftwich, 2010). In this study, participants appeared to negotiate this shift cautiously, maintaining the importance of human interaction, relational teaching, and contextual understanding.

From a TAM perspective, these concerns can be interpreted as factors influencing perceived ease of use and behavioral intention. While participants recognize the usefulness of AI, uncertainties regarding ethical use, accuracy, and pedagogical application may reduce their confidence in using AI effectively in teaching contexts. Extended TAM models emphasize that such concerns, particularly those related to risk and uncertainty, can act as barriers to adoption if not addressed through training and support (Venkatesh et al., 2003).

Importantly, the coexistence of positive perceptions and critical concerns reflects what can be described as a form of informed or cautious acceptance of AI. Rather than demonstrating resistance, participants exhibit a willingness to engage with AI, tempered by an awareness of its limitations and implications. This aligns with Mnguni's (2024) assertion that pre-service

teachers' perceptions of AI are often complex, context-dependent, and shaped by both optimism and uncertainty.

In the context of Initial Teacher Education, these findings suggest that perceptions of AI are not fixed, but are actively shaped by exposure, training, and contextual experience. As highlighted in Chapter 2, teacher education programmes play a crucial role in shaping how pre-service teachers interpret and evaluate emerging technologies (Aslan & Zhu, 2016; Viberg et al., 2023). Where AI is not meaningfully integrated into the curriculum, perceptions may remain superficial, fragmented, or influenced by informal experiences rather than critical pedagogical engagement.

The findings indicate that pre-service Life Sciences teachers demonstrate positive yet critically reflective perceptions of AI, characterized by strong perceived usefulness alongside concerns related to cognitive, ethical, and professional implications. These perceptions provide an important foundation for future AI integration; however, their translation into practice depends on the extent to which teacher education programmes can support the development of informed, critical, and pedagogically grounded engagement with AI.

5.4 Discussion of RQ3: Factors Shaping Preparedness for AI Integration

The findings of this study demonstrate that pre-service teachers' preparedness for AI integration is not determined solely by individual knowledge or perceptions, but is significantly shaped by institutional and contextual factors within ITE. These factors include limited formal training, curriculum gaps, infrastructural constraints, and broader systemic inequalities, all of which interact to influence how pre-service teachers develop preparedness for AI integration.

A central issue emerging from the findings is the lack of structured AI integration within teacher education programmes. Participants consistently indicated that, while they had been exposed to general digital technologies, AI had not been meaningfully incorporated into their training, particularly within subject-specific methodology courses. This aligns with existing literature suggesting that teacher education programmes often prioritize general ICT skills over emerging technologies such as AI, resulting in fragmented or insufficient preparation (Venketsamy & Hu, 2022; Shilenge & Ramaila, 2020; Mnguni, 2023). In the absence of

structured engagement, pre-service teachers are left to develop AI-related knowledge independently, which limits opportunities for pedagogical modelling and critical reflection.

From a AI-TPACK perspective, this lack of institutional support constrains the development of integrated knowledge structures. While participants may acquire elements of technological knowledge through informal exposure, the absence of guided opportunities to connect this knowledge with pedagogy and Life Sciences content inhibits the formation of TPACK. As Mishra and Koehler (2006) emphasize, the development of TPACK requires deliberate instructional design, where technologies are embedded within pedagogical practices and disciplinary contexts. In this study, the limited presence of AI within the ITE curriculum suggests that such integration has not yet been fully realized.

The findings also highlight the influence of contextual constraints, particularly within the South African educational landscape. Participants referred to challenges such as unreliable internet access, limited availability of digital devices, and the impact of load shedding, all of which shape their perceptions of what is feasible in classroom practice. These constraints reflect broader structural inequalities that have been widely documented in South African education (Jere et al., 2023; Tarisayi, 2024). Importantly, these contextual realities do not only affect access to technology but also shape pre-service teachers' expectations and confidence regarding future implementation.

Interpreted through the TAM, these contextual and institutional factors can be understood as external variables that influence both perceived ease of use and actual technology adoption (Venkatesh et al., 2003). While participants in this study generally perceived AI as useful, their ability to envision its practical application was often constrained by concerns about access, infrastructure, and support. This suggests that positive attitudes toward AI are insufficient for meaningful integration when enabling conditions are absent. Instead, preparedness emerges as a product of both individual disposition and environmental support.

Another significant factor shaping preparedness is the absence of practical, hands-on training opportunities within the ITE programme. Participants expressed a need for workshops, demonstrations, and subject-specific training that would allow them to experiment with AI tools in authentic teaching contexts. This aligns with research emphasizing the importance of

experiential learning in developing technological competence, particularly in relation to emerging technologies (Aslan & Zhu, 2016; Howard et al., 2022). Without such opportunities, pre-service teachers may develop theoretical awareness of AI without the confidence or competence to apply it in practice.

These findings also point to a broader issue of curriculum alignment within teacher education. The absence of AI-focused content within Life Sciences methodology suggests a disconnect between technological developments and curriculum design within ITE. As highlighted in Chapter 2, teacher education plays a critical role in shaping not only what teachers know, but how they think about teaching and learning (Ertmer & Ottenbreit-Leftwich, 2010; Mnguni, 2023). Where emerging technologies are not meaningfully integrated into the curriculum, pre-service teachers may perceive them as peripheral rather than integral to teaching practice.
Importantly, participants' responses suggest that their limited preparedness is not perceived as an individual failure, but rather as a reflection of systemic and institutional limitations. This reinforces the argument that preparedness for AI integration must be understood within a broader ecological framework, where individual, institutional, and contextual factors interact. Such an understanding shifts the focus from teacher deficit to the conditions under which preparedness is constructed and constrained.

At the same time, the findings reveal that participants are beginning to develop forward-looking pedagogical intentions, particularly in relation to the use of AI for simulations, differentiation, and the teaching of complex Life Sciences concepts. This suggests that, despite existing constraints, pre-service teachers are not resistant to AI integration but are instead navigating its possibilities within the limits of their current preparation and context. Their anticipated uses of AI reflect an emerging awareness of its pedagogical potential, even in the absence of formal training.

This positions preparedness as a dynamic and evolving construct, shaped by both current experiences within ITE and anticipated future teaching contexts. It also highlights the importance of aligning teacher education programmes with technological developments, ensuring that pre-service teachers are not only exposed to AI, but are supported in developing the knowledge, skills, and critical awareness required for its responsible and effective use.

5.5 Implications of the Study

The findings of this study have important implications for multiple levels of the education system, particularly within the ITE, but also extending to educational policy and future classroom practice. While the primary contribution of this study lies in understanding how AI preparedness is constructed within pre-service teacher preparation, the results also highlight broader systemic considerations that influence how such preparedness can be supported and sustained beyond the university context.

As demonstrated in Chapter 4 and discussed in this chapter, pre-service teachers' preparedness for AI integration is shaped not only by their individual knowledge and perceptions, but also by institutional structures, curriculum design, and contextual realities. This suggests that effective integration of AI in education requires alignment across different levels of the system, including teacher education programmes, policy frameworks, and school environments. The following sections outline the implications of this study across these interconnected levels, beginning with ITE as the primary focus, followed by broader policy and classroom-level considerations.

5.5.1 Implications for Initial Teacher Education

The findings of this study carry important implications for ITE, particularly in relation to how pre-service teachers are prepared for the integration of emerging technologies such as AI within subject-specific contexts. As demonstrated in Chapter 4, pre-service teachers in this study exhibit positive perceptions of AI and a willingness to engage with its pedagogical potential; however, their preparedness remains constrained by limited pedagogical training, fragmented knowledge development, and contextual challenges. These findings suggest a need for a more deliberate and structured approach to AI integration within teacher education programmes.

A key implication relates to the integration of AI within the ITE curriculum, particularly within subject methodology courses such as Life Sciences education. The findings indicate that current exposure to AI is largely informal and self-directed, with limited opportunities for guided engagement within formal coursework. This suggests that AI should not be treated as an optional or peripheral topic, but rather as an integral component of teacher preparation. Embedding AI within methodology modules would enable pre-service teachers to explore its

application in relation to specific disciplinary concepts, pedagogical strategies, and curriculum requirements. Such integration would support the development of TPACK, allowing pre-service teachers to move beyond basic technological familiarity toward meaningful pedagogical use (Mishra & Koehler, 2006).

Closely related to this is the need for practical, hands-on training opportunities that allow pre-service teachers to experiment with AI tools in authentic teaching contexts. The findings highlight a clear demand for workshops, demonstrations, and applied learning experiences where students can engage with AI in lesson planning, assessment design, and classroom simulation. Research suggests that experiential learning plays a critical role in developing both technological competence and confidence (Aslan & Zhu, 2016; Howard et al., 2022). Within the ITE context, such opportunities would enable pre-service teachers to develop not only technical skills, but also critical awareness of how AI can be adapted to diverse classroom environments.

The findings also point to the importance of incorporating ethical and critical AI literacy within teacher education. Participants expressed concerns related to accuracy, academic integrity, and over-reliance on AI, indicating an awareness of the ethical complexities associated with its use. This suggests that ITE programmes should move beyond technical training to include critical engagement with issues such as algorithmic bias, data privacy, and the responsible use of generative AI in teaching and learning (Ng et al., 2021; Howard et al., 2022). Developing such competencies is essential for preparing teachers who are not only users of AI, but also critical mediators of technology in the classroom.

Another important implication concerns the role of institutional support and curriculum alignment. The findings suggest that the absence of AI-focused content within teacher education contributes to the persistence of the theory–practice gap identified in this study. Addressing this gap requires curriculum reform that aligns technological developments with pedagogical training. This may include the introduction of dedicated AI modules, the integration of AI across existing courses, or the incorporation of AI-related competencies within teaching practice requirements. Such reforms would ensure that AI is positioned as a legitimate and expected component of professional teaching practice.

In addition, the study highlights the need for ITE programmes to be responsive to contextual realities, particularly within the South African educational landscape. Issues such as unequal access to resources, unreliable infrastructure, and the digital divide continue to shape what is feasible in classroom practice. Teacher education programmes must therefore prepare pre-service teachers not only to use AI in ideal conditions, but also to adapt its use within resource-constrained environments. This may involve exploring low-data AI tools, offline resources, or blended approaches that take contextual limitations into account. As Jere et al. (2023) and Tarisayi (2024) argue, context-sensitive approaches are essential for ensuring equitable and sustainable technology integration.

Furthermore, the findings suggest that teacher educators themselves play a critical role in shaping pre-service teachers' preparedness for AI integration. As highlighted in the literature, lecturers serve as key mediators of technological knowledge, influencing how students perceive and engage with emerging technologies (Ertmer & Ottenbreit-Leftwich, 2010). Where teacher educators demonstrate the pedagogical use of AI, integrate it into coursework, and create opportunities for critical discussion, pre-service teachers are more likely to develop informed and confident approaches to its use. This underscores the importance of professional development for teacher educators, ensuring that they are equipped to guide students in navigating AI in education.

Taken together, these implications suggest that preparing pre-service teachers for AI integration requires a systemic and multi-dimensional approach, encompassing curriculum design, pedagogical practice, ethical engagement, and contextual responsiveness. Rather than positioning AI as an additional skill, teacher education programmes must recognize it as part of the evolving landscape of teaching and learning, requiring deliberate integration into the professional formation of future educators.

5.5.2 Implications for Policy

Beyond the context of Initial Teacher Education, the findings of this study highlight important considerations for educational policy, particularly in relation to the integration of artificial intelligence within teaching and learning. The identified gaps in pre-service teachers' preparedness suggest that AI integration requires clearer strategic direction at a systemic level,

rather than reliance on informal or institution-specific practices. This includes the development of policy guidelines that address not only the use of AI tools in classrooms, but also issues related to ethical use, assessment integrity, and data privacy (UNESCO, 2021; Holmes et al., 2019).

In the South African context, these implications are closely linked to broader efforts toward digital transformation. However, structural inequalities such as uneven access to infrastructure, unreliable connectivity, and resource disparities continue to shape what is feasible in practice (Jere et al., 2023; Tarisayi, 2024). Policy responses must therefore move beyond aspirational goals and address these material conditions to support equitable AI integration.

There is also a need for stronger alignment between policy expectations and teacher preparation. Ensuring that AI-related competencies are reflected within teacher education frameworks can support a more coherent transition from training to practice, bridging the gap between national goals and classroom realities (Selwyn, 2016).

5.5.3 Implications for School-Level Practice and Future Classrooms

At the level of classroom practice, the findings suggest that the integration of AI will depend not only on teachers' preparedness, but also on the conditions within which they work. As pre-service teachers transition into professional roles, their ability to implement AI meaningfully will be influenced by school infrastructure, institutional support, and access to appropriate technological resources.

The findings also point to potential implications for teaching practice and learner outcomes. Participants recognised the value of AI in supporting the teaching of complex Life Sciences concepts through visualization, simulation, and differentiation. When used appropriately, such tools may enhance conceptual understanding and support diverse learning needs (Luckin et al., 2016; Holmes et al., 2019). However, these benefits require careful pedagogical use to avoid over-reliance and to maintain critical thinking and learner engagement (Selwyn, 2016).
The role of the teacher therefore remains central. Rather than replacing teaching, AI is likely to reshape instructional practices, requiring teachers to act as facilitators and critical evaluators of knowledge (Ertmer & Ottenbreit-Leftwich, 2010). This highlights that effective AI

integration in future classrooms will depend not only on access to technology, but on how teachers apply it in pedagogically meaningful and contextually responsive ways.

5.6 Chapter Summary

This chapter has discussed the findings of the study in relation to the research questions and the theoretical frameworks guiding the study, namely TAM and AIPK. The discussion highlighted that while pre-service Life Sciences teachers recognize the pedagogical potential of artificial intelligence, particularly for enhancing conceptual understanding, their preparedness for integration remains uneven due to limited pedagogical training and contextual constraints.

The findings further suggest that positive perceptions of AI, reflected in its perceived usefulness, are not matched by sufficient perceived ease of use in classroom contexts, largely due to a lack of structured preparation within ITE. The chapter has therefore positioned preparedness for AI integration as shaped by the interplay between knowledge, training, and contextual realities, providing a foundation for the concluding chapter, which outlines the overall contributions, limitations, and recommendations of the study.

CHAPTER SIX

RECOMMENDATIONS AND CONCLUSION

6.1 Introduction

Drawing directly from the thematic findings presented in Chapters 4 and 5, the chapter synthesises the key conclusions, outlines the study's limitations, and presents targeted recommendations for universities, the DBE, and future research. This study set out to explore Life Sciences pre-service teachers' perceptions and preparedness regarding the integration of AI in their future classrooms. The findings reveal a cohort that is intellectually open and pedagogically optimistic about AI yet professionally underprepared for structured classroom implementation.

Participants consistently recognised AI as a powerful pedagogical tool capable of enhancing the teaching of abstract and microscopic biological processes. They associated AI with visualisation, simulation, and automated formative assessment, suggesting a clear awareness of its potential to scaffold complex Life Sciences content. At the same time, they did not perceive AI as a replacement for the teacher, but rather as a supportive tool that enhances facilitation, learner engagement, and efficiency.

However, despite this positive orientation, a significant gap emerged between personal exposure to AI tools and formal pedagogical training for subject-specific integration. Participants reported that AI was mentioned informally within their programme but not embedded in a structured or practical manner. This has resulted in moderate confidence rather than professional preparedness.

The findings further demonstrate that perceptions of AI integration are shaped by the structural realities of South African schooling. Concerns regarding electricity, internet access, and unequal resource distribution strongly influence anticipated implementation. Ethical considerations, including learner dependency, misinformation, and the need to verify AI-generated content against CAPS requirements, also temper enthusiasm with caution.

6.2 Summary of Key Findings

This study sought to explore pre-service Life Sciences teachers' preparedness, perceptions, and preparedness for the integration of AI within the context of Initial Teacher Education. The findings reveal that AI preparedness is not a singular construct, but a multi-dimensional phenomenon shaped by knowledge, perceptions, and contextual factors.

In relation to RQ1, the findings indicate that pre-service teachers demonstrate basic but uneven preparedness for AI integration. While participants showed familiarity with AI tools and expressed confidence in using them for personal and academic purposes, their knowledge was largely informal and self-directed**,** rather than developed through structured teacher education. A clear distinction emerged between general technological awareness and pedagogical application, with participants indicating uncertainty about how to integrate AI meaningfully into Life Sciences teaching. This reflects a significant theory–practice gap**,** where conceptual awareness of AI is not matched by pedagogical competence.

Regarding RQ2, the findings show that pre-service teachers hold generally positive yet critically balanced perceptions of AI in Life Sciences education. Participants viewed AI as a valuable pedagogical support tool, particularly for enhancing the teaching of abstract and complex biological concepts through visualisation and simulation. At the same time, these positive perceptions were accompanied by concerns related to cognitive dependency, reduced critical thinking, academic integrity, and ethical use**. P**articipants also expressed awareness of their professional role in mediating AI use, emphasising the importance of maintaining pedagogical judgement and preserving the human dimensions of teaching.

In relation to RQ3, the findings indicate that preparedness for AI integration is shaped by both institutional and contextual factors**.** Participants highlighted the absence of structured AI training within their teacher education programme, as well as broader challenges such as limited infrastructure, unreliable internet access, and unequal distribution of technological resources. Despite these constraints, participants demonstrated emerging awareness of how AI could be applied in teaching, particularly in relation to simulations, differentiation, and assessment support.

Taken together, the findings suggest that while pre-service teachers are receptive to AI and recognise its pedagogical potential, their preparedness for meaningful integration remains constrained by limited pedagogical preparation within ITE and broader contextual realities.

6.3 Recommendations

The recommendations presented below are grounded directly in the findings of this study and respond to the identified gap between positive attitudes toward AI and limited structured preparation for its integration.

6.3.1 Recommendations for Universities and Teacher Education Institutions

The most consistent recommendation emerging from participants was the need for structured and formal training in AI integration. Participants repeatedly indicated that AI was only mentioned briefly during lectures and not systematically embedded within their programme. This suggests that incidental exposure is insufficient for developing pedagogical preparedness.

Universities should therefore embed AI integration within subject-specific methodology modules, particularly in Life Sciences pedagogy courses. Rather than treating AI as a general technological tool, training should focus on how AI can support the teaching of specific biological concepts, such as cellular processes, genetics, and ecological systems. This would directly address the knowledge-practice gap identified in the findings.

In addition, institutions should consider introducing a dedicated, credit-bearing module or structured short course focused specifically on AI in education. Participants expressed a clear desire for formal preparation that goes beyond theoretical awareness. Such a module should include training on evaluating the scientific accuracy of AI-generated content, aligning AI outputs with CAPS requirements, ethical classroom use, and designing AI-supported assessments.

Importantly, some participants suggested that structured AI training could culminate in a certificate of competence in AI integration. Providing certification would not only acknowledge acquired skills but also enhance professional confidence and accountability. Formal recognition may also strengthen graduates' employability in increasingly digital school environments.

Finally, practical, hands-on workshops should form a central component of AI preparation. Confidence develops through practice rather than theoretical discussion. Simulated teaching scenarios, lesson design tasks, and guided evaluation of AI-generated Life Sciences materials

would provide pre-service teachers with experiential preparation before entering real classrooms.

6.3.2 Recommendations for the Department of Basic Education

The successful integration of AI in Life Sciences classrooms cannot depend solely on individual teacher preparedness. The findings of this study demonstrate that contextual and infrastructural realities significantly shape the perceived feasibility of AI use. Therefore, systemic support from the Department of Basic Education is essential.

Firstly, infrastructural development must be prioritised to ensure equitable digital access across schools. Participants expressed concern that unstable electricity supply, limited internet connectivity, and unequal access to devices may hinder AI integration and potentially widen the gap between well-resourced urban schools and under-resourced rural schools. Without addressing these foundational barriers, AI integration risks reinforcing existing inequalities rather than reducing them.

Secondly, the DBE should explore the procurement or development of low-data or offline-capable AI tools that can function in contexts affected by load shedding and connectivity challenges. Context-sensitive technological solutions are necessary for meaningful implementation within South African schooling environments.

Thirdly, structured and subject-specific professional development should be made available to in-service teachers. As indicated by participants, many teachers currently in the field may not have received formal training in AI integration. Continuous professional development programmes, particularly those aligned with Life Sciences pedagogy and CAPS requirements, would support ethical and effective classroom use. Establishing communities of practice could further allow teachers to collaboratively evaluate AI-generated content, share lesson resources, and develop responsible integration strategies. AI integration at school level therefore requires coordinated institutional action, infrastructural investment, and sustained professional support rather than isolated individual effort.

6.4 Limitations of the Study

This study employed a qualitative exploratory case study design within a single higher education context. The sample consisted of 38 questionnaire responses and four interview

participants, which allowed for in-depth exploration of pre-service teachers' perceptions and preparedness. However, the findings are context-specific and cannot be statistically generalized to all Life Sciences pre-service teachers in South Africa.

The study relied on self-reported data, which reflects participants' perceptions, experiences, and anticipated practices rather than observed classroom implementation. As a result, the findings provide insight into how pre-service teachers understand and position themselves in relation to AI, but do not capture how these understandings translate into actual teaching practice. Future research could adopt longitudinal or classroom-based approaches to examine how AI integration unfolds in real teaching environments.

In addition, the research was conducted during a period of rapid technological development, where AI tools and their educational applications continue to evolve. Participants' perceptions of AI may therefore shift over time as technologies become more advanced, accessible, and embedded within educational systems. The findings should thus be understood as reflecting a particular moment in the ongoing development of AI in education.

Finally, the researcher's positionality as a participant within a similar educational context may have influenced both data collection and interpretation. While this shared context facilitated rapport and openness during data generation, reflexive awareness was maintained throughout the research process to minimize potential bias and ensure credibility in the analysis.

6.5 Conclusion

This study examined pre-service Life Sciences teachers' perceptions and preparedness for the integration of artificial intelligence within the context of Initial Teacher Education. The findings reveal a cohort of future educators who recognize the pedagogical potential of AI, particularly for enhancing the teaching of complex and abstract biological concepts. Participants consistently positioned AI as a supportive tool that can improve efficiency, facilitate understanding, and enrich classroom practice, rather than as a replacement for the teacher.

At the same time, the study highlights a clear distinction between positive perceptions and professional preparedness. While participants demonstrated awareness of AI tools and

expressed willingness to integrate them into their future teaching, their knowledge was largely informal and self-directed. The absence of structured, subject-specific training within teacher education programmes contributed to moderate levels of confidence and uncertainty about pedagogical application. In addition, contextual challenges such as infrastructural inequality and ethical concerns further shaped how preparedness for AI integration was understood.

The study contributes to the growing body of research on AI in education by foregrounding the importance of ITE on as a critical site for developing AI preparedness. It demonstrates that preparedness is not determined solely by access to technology or positive attitudes, but by the extent to which teacher education programmes provide opportunities for pedagogical integration, critical engagement, and context-sensitive application.

Ultimately, the integration of AI in Life Sciences education will depend not only on technological advancement, but on the preparation of teachers who are able to use these tools in ways that are pedagogically meaningful, ethically responsible, and responsive to diverse classroom contexts. This study therefore underscores the need for deliberate and structured approaches to AI integration within teacher education, ensuring that future educators are equipped to navigate the opportunities and challenges of an evolving educational landscape.

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APPENDIX A

Demographics Table

Participant	Gender	Age Range	Prior AI Exposure	Description of Use
P1	Female	20–22	Limited	Basic use for meanings and simple explanations
P2	Female	20–22	Moderate	Studies, teaching practice, personal use
P3	Female	22–25	Limited	Basic tools (ChatGPT, Meta AI)
P4	Female	22–25	Moderate	Lesson resources and simplification
P5	Female	22–25	Broad	Multiple tools (Copilot, DeepSeek, etc.)
P6	Female	25–30	Moderate	Studies and teaching preparation
P7	Male	25–30	Moderate	Observed AI in teaching contexts
P8	Female	22–25	Broad	Paraphrasing, summarising, structuring
P9	Female	20–22	Moderate	ChatGPT and DeepSeek
P10	Male	20–22	Moderate	Academic writing support
P11	Male	20–22	Moderate	Assignments and concept understanding
P12	Female	20–22	Moderate	Frequent use across contexts
P13	Female	20–22	Moderate	Teaching strategies and ideas
P14	Female	20–22	Moderate	Studies and learner observation
P15	Female	20–22	Limited	Occasional lesson examples
P16	Female	20–22	None	No prior exposure
P17	Male	20–22	Moderate	Content understanding and advice
P18	Female	20–22	Broad	ChatGPT, Copilot, Grammarly, Kahoot
P19	Female	20–22	Moderate	Writing and brainstorming
P20	Male	20–22	Limited	Minimal use
P21	Female	22–25	Moderate	Lesson modification and concepts
P22	Female	20–22	Moderate	Meta AI and ChatGPT
P23	Female	20–22	Limited	Simplifying lecture content
P24	Female	19	Moderate	ChatGPT and MetaAI
P25	Female	20–22	Moderate	Grammar correction and feedback
P26	Female	20–22	Moderate	Copilot and ChatGPT
P27	Female	19	Moderate	Academic and personal use
P28	Female	20–22	Broad	ChatGPT, Teachy, PhET
P29	Female	20–22	Moderate	Clarification and learner misuse
P30	Female	22–25	Moderate	Copilot and Gemini
P31	Male	22–25	Limited	ChatGPT, Meta AI, Notebook AI
P32	Male	20–22	Broad	Conferences and teaching use
P33	Male	20–22	Limited	NotebookLM and DeepSeek
P34	Male	20–22	None	No exposure
P35	Male	20–22	Broad	Lesson planning and daily tasks
P36	Male	20–22	Broad	Canva, Quizizz, ChatGPT
P37	Male	22–25	Broad	Academic, teaching, personal use
P38	Female	20-22	Limited	Studying, ChatGPT

APPENDIX B

Questionnaire

Section 1: Participant Information and Consent

Participants were informed about the purpose of the study, voluntary participation, confidentiality, and ethical approval.

Consent Question:

I have read and understood the information provided above. I voluntarily agree to participate in this research study.

- Yes
- No

Section 2: Background Information

1. Gender:
   - Female
   - Male
   - Other: _____
2. Age:
   - 20–22
   - 22–25
   - 25–30
   - Other: _____
3. What is your teaching major?
4. Have you completed school teaching practice? If yes, briefly describe the type of school.
5. Have you encountered AI tools in your studies, teaching practice, or personal life? Give examples.

Section 3: Perceptions of AI in Education

1. What does 'Artificial Intelligence in education' mean to you?
2. How could AI be useful in Life Sciences classrooms?
3. What opportunities or benefits might AI bring?
4. What concerns or risks do you associate with AI?
5. Will AI change the role of teachers? How?

Section 4: Anticipated Use of AI

1. How would you use AI to teach difficult Life Sciences concepts?
2. Which AI tools would be most relevant? Why?
3. How should AI support you as a teacher?

Section 5: Preparedness and Knowledge

1. How confident are you in integrating AI into teaching?
2. Have you received training on AI?
3. What support is needed to use AI effectively?

Section 6: Contextual Factors

1. What digital or AI tools did you observe during teaching practice?
2. How do infrastructure and resources affect AI use?

Section 7: Final Reflections

1. Do you intend to use AI in your teaching? Why?
2. What recommendations would you give for AI integration?

APPENDIX C

Semi-Structured Interviews

Opening

Introduction of the researcher, explanation of the study, assurance of confidentiality, and request for consent to record.

Main Questions

1. What does AI mean to you in Life Sciences teaching?
2. How would you use AI to teach a Life Sciences topic?
3. How prepared do you feel to integrate AI?
4. What training experiences influenced your preparedness?
5. What opportunities do you see for AI?
6. What concerns do you have about AI?
7. How might AI affect the role of teachers?
8. What support is needed for AI integration?

Probes

• Can you give an example?

• Can you explain further?

• Why do you think that?

• How would that work in your classroom?

Closing

Any additional comments. Thanking participant.

APPENDIX D

Example of a Transcript Extract

Research Participant 1-20260130_104344-Meeting Recording

January 30, 2026, 3:43PM

29m 18s

Thembisile Buthelezi started transcription

TB **Thembisile Buthelezi** 0:12

Good day and thank you for agreeing to participate in this interview. My name is Thembisile Buthelezi, and I'm doing my masters at the University of the Witwatersrand. I'm conducting this research as part of my study, which is titled Life Sciences Pre-service Teachers, Perceptions and Preparedness for the Integration of AI in One South African University. So the purpose of this interview is to explore your views and experiences regarding AI in life sciences education. There are no wrong or right answers. I'm just interested in your honest opinions and experiences. Your participation in this study is completely voluntary. You may choose not to answer any questions and you may withdraw from the interview at any time without any penalties.

This interview will take approximately 20 to 30 minutes. With your permission, I would like to record the interview to ensure that your responses are captured accurately. The recording will be used for my research purposes only and will be kept confidential.

Your identity will not be disclosed in any report or publications arising from this study. All responses will be anonymized and treated with strict confidentiality. So do I have the right to record?

NM **Participant 1** 1:48

Yes, you do.

TB **Thembisile Buthelezi** 1:50

Thank you. Do you have any questions before we begin?

NM **Participant 1** 1:54

No, I don't have any questions. It is clear.

TB **Thembisile Buthelezi** 1:58

Do you contend to proceed with this interview?

NM **Participant 1** 2:06

Yes, I am.

Thembisile Buthelezi 2:09

Thank you, I am going to start with my first question. So what does AI mean to you in the context of teaching life sciences?

Participant 1 2:29

OK, so to me, artificial intelligence in the context of teaching life sciences is not really about replacing the teaching or making the learning automatic. But instead, I see it as a support tool, like something that can help both teachers and learners understand the complex biological concepts in more meaningful and accessible ways.

Like life sciences, as we know that life sciences is a subject that deals with many aspects of ideas, so maybe like cellular processes, genetic evolution, etc. So these concepts are often difficult for learners.

So to visualize or relate to their everyday experience, we can think of AI and think of the tools that can break down these concepts, providing our virtual simulation, generate explanations at different levels and even respond to learners' questions in real time. So I also see AI as something that can help with personalized learning in a typical life sciences classroom.

Learners can have very different abilities, background and a learning speed or to understand something. So AI can help cater for those differences by offering extra explanations and also to use AI to practice with difficult questions or any question that they do not understand and also to provide the examples that the teacher didn't provide in the classroom with that maybe the learner feeling embarrassed because some of the learners are too shy to ask in the classroom. So they may like, use AI to ask questions and provide examples without feeling embarrassed to ask the teacher like so often. However, I don't see AI as something neutral or automatically good. For instance, in the South African context, AI is deeply connected to issues of access, inequity and ethics.

So for me, artificial intelligence in life sciences teaching is both an opportunity and challenge. It has the potential to enhance learning, but only if it is used thoughtfully and in a right manner and in a way that can make sense for our learners and teachers, yeah.

Thembisile Buthelezi 5:41

OK. OK. Thank you very much. Thank you very much for the thoughtful answer. My next question will be asking you how you would use artificial intelligence when teaching a basic lab sciences topic and I'm going to touch on your response. You mentioned examples of teaching genetics or evolution as abstract topics. And you also mentioned that you can use AI enhanced visuals to explain those topics. But just for the sake of this second question, I can see that you've already answered.

But I wanted to just get a standalone answer so that I can get your clear perspective on this. So exactly how would you use AI when you are teaching one of those life sciences topics? You can just choose one topic and.

Give any idea that comes to mind on how you would use AI specifically to make those abstract topics easier as you said it and end the other question.

Participant 1 6:51

Well, let us take maybe the DNA structure and the protein synthesis. The topic of the DNA structure and the protein synthesis like many learners struggle with.

NM I can see several ways of using AI. For example, I can use AI powered Simulation and maybe animation. Is it animation? I don't know whether I'm pronouncing it correctly. That shows how the DNA replicates or how the transcription or translation takes place. So instead of learners just memorizing steps from the textbook they could actually see the process unfolding. They can pause it and replay it and ask maybe questions about the specific stages if they have any questions or if they do not understand somewhere and I could also use the AI to generate the simplified explanation of the same concept at the different level you see maybe for example, some learners may need like the very basic explanation in order for them to easily understand because when you give them maybe...I don't know maybe some difficult explanation they may not understand so they may want you to maybe to break it down a little bit in order for them to understand. So, I can use AI for some learners that may need the very basic explanation using maybe the everyday analogies, while others might be ready for more scientific language. So, AI could help me provide both explanations without taking up too much time in the classroom and also another way I will use maybe for assessment and revision where artificial intelligence can generate the questions, the quizzes, and maybe how I can maybe style my exam questions, obviously aligned with the kids document and learners could get like maybe the immediate feedback, which is something we often struggle to provide in in like overcrowded classes.

And this feedback can help them identify where they are going wrong before maybe like the formal assessment or exams.

And however, I will still guide the process of using the AI. I wouldn't just tell the learners to see; I wouldn't just tell the learners to ask AI and just leave them alone. I will teach them how to question information, how to check the accuracy, how to connect the AI explanation back to the curriculum and the real-life biological examples.

Thembisile Buthelezi 10:22

TB OK. OK. Thank you very much. And my next question is based more on your teaching practice, on your teaching, on your teaching practice and I'm assuming you have done teaching practice before, right?

Participant 1 10:39

NM Yes, I have.

Thembisile Buthelezi 10:40

TB OK, OK, so what experiences during your teacher training, which is in this case referring to your practice, have influenced how prepared or unprepared you feel to integrate AI into your teaching. So based on the teacher practice experience that you've had, do you feel like you are prepared to use AI or you are not prepared and how did that teaching experience influence that decision whether that preparedness if you are prepared or that unpreparedness if you are not prepared?

Participant 1 11:22
Could you please repeat your question?

Thembisile Buthelezi 11:25
OK, in simple terms, I'm asking during your teaching practice as a student, how did that practice affect you being prepared to use AI in your future classrooms or if you are not being prepared to use AI in your future classrooms?
Do you want me to also simplify it again because I can start with asking, maybe do you think you are prepared to use AI in your life sciences classrooms?

Participant 1 11:56
OK, um.
I understand your question now. So to honestly answer this question, my teaching training has made me feel partially prepared and most uncertain about the integration of AI into teaching. On the other hand, I've been exposed to discussions about technology in education and online learning platforms, digital resources, especially after the COVID-19. That exposure helped me realize that technology is no longer optional in teaching but a must. However, when it comes to artificial intelligence specifically, the training has been very limited. We were not really shown how AI works.
How to use it responsibly or how to integrate it meaningfully into a science lesson or into the classroom environment. So most of what I've known about AI has come from my own exploration, social media or informal discussion with my peers or friends. So during the teaching practicals, the schools themselves have influenced how prepared I felt. Some schools had smart boards and Internet access, while others had none of that, especially in the rural or township schools. So this made it difficult to practice using the AI tools in the real classroom. So even if we discuss the innovation at the university, the reality at the schools often pushes us back to the traditional like the chalk and board so um in overall, my training helped me become aware. Oh yeah, my training helped me become aware of AI, but not necessarily confident in using AI.

Thembisile Buthelezi 14:44
Mm.
OK, OK.

Participant 1 14:48
So I feel like there is a gap between theory and practice when it comes to AI integration, especially at that. I feel like they're prioritizing theory most of the time rather than the practice or practicals.

Thembisile Buthelezi 15:07
OK, I agree, I agree. My next question will be asking... are you confident or how confident are you about integrating AI responsibly and effectively in your life sciences classroom and why? But I want to note that in your previous answer you did mention that you are not confident

enough because you feel like you are only aware of AI but you have not been trained on how to use it. So just for the sake of this question I've got that answer but I would like it if you can also answer this one and also follow it with a clear why. So how confident do you feel about using AI responsibly and effectively in your life sciences classroom and why?

TB

Participant 1 16:06

So as I stated that my confidence level in using AI is just moderate. It's in between moderates. So I will put the confidence in my ability to learn and adapt. But not fully confident that I am ready to have all the skills needed to use the AI responsibly and effectively. So one reason for this is that AI can produce information that sounds correct but may not always be accurate or maybe align with the curriculum or which word is needed at that particular time. So as a life sciences teacher, I carry the responsibility of ensuring that the learners receive the correct and the curriculum aligned knowledge or information. That means I will need to double check what AI is generating which requires strong content knowledge and the hard thinking on my side.

APPENDIX E

Ethical Clearance

UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG | WSoE Wits School of Education

SCHOOL OF EDUCATION ETHICS COMMITTEE

CONSTITUTED UNDER THE UNIVERSITY HUMAN RESEARCH ETHICS COMMITTEE (NON-MEDICAL)

CLEARANCE CERTIFICATE	**PROTOCOL NUMBER: 2025ECEO70M**
PROJECT TITLE	
Life sciences pre-service teachers' perceptions and knowledge readiness for integration of AI in one South African university	
INVESTIGATOR	Buthelezi, Thembisile
SCHOOL/DEPARTMENT OF INVESTIGATOR	WSOE
DATE CONSIDERED	08 September 2025
DECISION OF THE COMMITTEE	Approved unconditionally.
RISK LEVEL	**Minimal Risk**
EXPIRY DATE	Date of submission of the Research Report
ISSUE DATE OF CERTIFICATE	26 September 2025
CHAIRPERSON	
	Dr. Peter Merisi

Supervisor: Dr. Mpunki Nakedi

DECLARATION OF INVESTIGATOR

To be completed in duplicate and **ONE COPY** returned to the Chairperson of the School/Department ethics committee.

I fully understand the conditions under which I am are authorized to carry out the abovementioned research and I guarantee to ensure compliance with these conditions. Should any departure to be contemplated from the research procedure as approved I/we undertake to resubmit the protocol to the Committee.

_____ **Signature**

_____/_____/_____ **Date**

Name of researcher: Thembisile Buthelezi (Student number: 1851549)

Title of project: Life Sciences Pre-Service Teachers' Perceptions and Preparedness for Integration of AI in one South African University

I, _____, agree to participate in this research project.

I agree to the following:

Statement	**YES**	**NO**
The research study was explained to me. I understand what this study is about.		
I understand that participation is voluntary, and I may withdraw at any time without penalty.		
I agree to complete the open-ended questionnaire.		
I agree that a follow-up semi-structured interview may be conducted with me.		
I agree that the semi-structured interview will be audio-recorded.		
I agree that direct quotations from my questionnaire or interview responses may be used in the research report, provided my identity is kept anonymous.		
I agree that my participation will remain anonymous (my name or other identifying details will not be used in the report).		
I agree that other researchers may use the anonymized data I provide (after obtaining their own ethics clearance), but my name and any personal information will not be shared.		

Participant's signature: _____

Name of participant (print): _____

Date: _____

Researcher's signature: _____

Name of researcher/person seeking consent: _____

Date:

_____

APPENDIX G

Permission Letter

UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG | WSoE Wits School of Education

27 October 2025

Thembisile Buthelezi
1851549@students.wits.ac.za

Re: Permission to conduct research at the Wits School of Education - PROTOCOL NUMBER: 2025ECEO70M

Dear Ms Buthelezi,

Permission is, hereby granted for you to conduct your study **"Life sciences pre-service teachers' perceptions and knowledge readiness for integration of AI in one South African university."** at the Wits School of Education in accordance with **PROTOCOL NUMBER: 2025ECEO70M** of your unconditional ethical clearance approval.

Sincerely,

JC Perumal

Professor Juliet Perumal
Head of School
Wits School of Education
Tel: 011 717 3003
Email: Juliet.Perumal@wits.ac.za

University of the Witwatersrand
School of Education
Participant Information Sheet

Researcher: Thembisile Buthelezi
Student Number: 1851549
Master of Education Candidate
1851549@students.wits.ac.za
20-08-2025

Dear student

RE: Invitation to Participate in my Master of Education Research Project

My name is Thembisile, a Master of Education student here in the Wits University, School of Education. I am conducting research as part of my degree requirements under the supervision of Dr Mpunki Nakedi and would like to invite you to participate in my study. Below I give you a full outline of the research project and what to expect if you accept my invitation.

Title of the Study: *Life Sciences Pre-Service Student Teachers' Perceptions and Preparedness for Integration of AI in one South African University*

1. Purpose of the Study

This study will explore the perceptions, preparedness, and anticipated practices of pre-service Life Sciences teachers in relation to integrating Artificial Intelligence (AI) into Life Sciences teaching. The findings will contribute to understanding how teacher education can better prepare future educators for AI-enhanced classrooms.

2. Procedures

If you agree to participate:

- You will complete an open-ended questionnaire, either online or in hard copy, taking approximately 20–30 minutes.
- A small number of participants (4–6) may be invited to take part in a follow-up semi-structured interview lasting about 30–40 minutes. Interviews will be conducted either in person or via an online platform such as Microsoft Teams or WhatsApp (audio only), depending on your preference.
- With your consent, interviews will be audio-recorded to ensure accuracy in capturing your views.

3. Voluntary Participation

Your participation is voluntary. You may choose not to answer certain questions or withdraw from the study at any point without penalty. Choosing not to participate will not affect your academic standing or relationship with the University.

4. Risks and Benefits

This study poses no known risks. While there are no direct benefits to you, your participation will help shape knowledge about teacher preparation for AI integration in Life Sciences classrooms, potentially influencing teacher education practices.

5. Confidentiality

All information you provide will be treated confidentially.

- Your name will not appear in any reports; pseudonyms will be used.
- Data will be stored securely on a password-protected device.

- Only the researcher and academic supervisors will have access to the data.
- Audio recordings will be destroyed once they have been transcribed and checked.

6. Contact Details
If you have questions about the study, please contact:
Thembisile Buthelezi 1851549@students.wits.ac.za
Cell: 0637473855
Supervisor: Dr Mpunki Nakedi
Division of Science and Technology Education
Tell: 011 717 3095

THANK YOU SO MUCH